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Geological Society of America Bulletin 2000;112;1315-1341
doi: 10.1130/0016-7606(2000)112<1315:BOTKID>2.0.CO;2

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Base of the Kiaman: Its definition and global stratigraphic significance

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ABSTRACT

Defining and dating the base of the Kiaman (the long period of reverse polarity that spans the time from the Late Carboniferous to the middle Permian) have proven difficult. In an attempt to do this an integrated stratigraphic, paleomagnetic, and geochronologic study was undertaken in Carboniferous rocks in the northern Tamworth Belt of the New England orogen, eastern Australia.

New paleomagnetic measurements at 145 horizons (sites) in the New England orogen of eastern Australia provide an accurate record of the late Namurian to Westphalian geomagnetic field. They have high unblocking temperatures, satisfy tilt, conglomerate, and slump tests, and show excellent agreement between a wide variety of igneous and sedimentary rock types, indicating that magnetism was acquired at, or soon after, deposition. Magnetite and, to a lesser degree, hematite are the carriers of magnetization. The record is contained in sequences of glaciogene sediments and arc-derived volcano-sedimentary units, folded into broad structures in the Late Permian to Triassic. New isotopic dates for the volcanic units indicate ages from 321 to 306 Ma (late Namurian to Westphalian). Magnetizations are steeply inclined downward (reversed polarity in the Southern Hemisphere) except within the Clifden Formation of the Rocky Creek syncline, where a change from normal to reverse polarity referred to as the Wanganui reversal records the onset of the Kiaman superchron.

Integrating complex stratigraphic, isotopic, and paleomagnetic data, we correlate the Wanganui reversal to the top of N6 in the Joggins section Nova Scotia, Canada, and estimate that its age is between 318 and 316 Ma. Further work may substantiate a younger normal zone in Australia that would support a slightly younger age, but within these limits. These considerations indicate that the Wanganui reversal occurs in the late Namurian (Marsdenian) and after the time of the Mississippian-Pennsylvania boundary. The mean direction of magnetization of the Upper Carboniferous Australian sequences is declination (D) = 195.3°, inclination (I) = 76.9°, $\kappa = 25$, $\alpha_{95} = 3.1°$, with a paleopole at 51.9°S, 141.2°E, $\alpha_{95} = 5.7°$, and a paleolatitude of 65.0° ± 5.6°S (P = 0.05). This paleolatitude is consistent with the glacial origin of some of the sedimentary units.

Keywords: Australia, Carboniferous, chronology, Kiaman, magnetostratigraphy, paleomagnetism, stratigraphy.

INTRODUCTION

Irving and Parry (1963), from their studies in Australia and paleomagnetic data from other continents, recognized that a very long period of reverse polarity of Earth's magnetic field existed during Permian-Carboniferous time. They named this unit the Kiaman magnetic interval, because the first observation from it of reverse polarity was from rocks (from the Blow Hole latite flow) at Kiama in New South Wales (Mercanton, 1926). Rather than the cumbersome term ''Permian-Carboniferous reversed superchron'' (Irving and Pullaiah, 1976), we refer to this large stratigraphic unit as the Kiaman reversed superchron, or simply the Kiaman, commemorating the original discovery.

Prior to the work of Irving and Parry, reversal sequences had been described mainly from Cenozoic rocks that indicated that the geomagnetic field reversed every million years or so (Hospers, 1954), and their work provided the first recognition that much longer polarity intervals (chrons) are present in the magnetic record. The Kiaman is the longest polarity chron known.

The geomagnetic field originates in the core and there has been much speculation about the significance of long intervals of constant polarity for the history of the core and mantle (Hide, 1967; Irving and Pullaiah, 1976; Courtillot and Besse, 1987; Larson, 1991). Locating and dating of reversals marking the bottom and top of the Kiaman are needed for the advancement of this discussion. Moreover, because reversals take place in a few thousand years, recognition of these horizons would provide global stratigraphic markers of very high accuracy. This is

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GSA Bulletin; September 2000; v. 112; no. 9; p. 1315–1341; 18 figures; 3 tables.

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especially important for a long-standing problem of correlating late Paleozoic rocks of Gondwana with those of Laurasia, a task made difficult by the lack of species common to both the cold climate Gondwanan biota and contemporaneous warm climate biota of the northern continents.

This study therefore has two main purposes. The first is to locate and date the sequence that records the change from normal to reverse polarity at the base of the Kiaman in the Carboniferous of eastern Australia. The second is to match this to a comparable sequence in North America (actually in the Joggins section of eastern Canada; DiVenere and Opdyke, 1991b), thus demonstrating the utility of this reversal as a global stratigraphic isotopic horizon. These tasks involve the application of complex stratigraphic, radiometric dating, paleomagnetic and paleontological methods, and the coherence that we have been able to achieve among them is a measure of their reliability. Our results also provide estimates of paleolatitude of Australia in the Late Carboniferous (Namurian to Westphalian).

Attempts to define limits to the Kiaman globally were reviewed by Opdyke and Channell (1996, p. 215–220). Irving and Parry (1963) defined the lower boundary in the Hunter Valley as being between the Paterson Volcanics, which have normal polarity, and overlying glacial varves of the Seaham Formation, which are reversed. However, U/Pb (sensitive high-resolution ion microprobe, SHRIMP) dates have subsequently shown that there is a gap of as much as 14 m.y. between these two levels, making them unsuitable for stratigraphic definition (Claoué-Long et al., 1995; Roberts et al., 1995a). Consequently, we have sampled what detailed field mapping (Brown, 1987a, 1987b; Stroud, 1990; Roberts, unpublished data) has shown to be more complete sections farther north in the Werrie and Rocky Creek synclines (Fig. 1). These sections were originally sampled by Irving (1966), but did not penetrate to the base of the Kiaman. His work showed that lithologies with well preserved magnetic records (ignimbrites, andesitic to dacitic flows, ash deposits, and glaciogene diamictites) are distributed throughout the thick succession. We have now extended sampling throughout the entire Visean–Late Carboniferous section in both the Werrie and Rocky Creek synclines, as well as the Kathrose–Maules Creek and Darthula blocks, which are located immediately west of the Rocky Creek syncline and are separated from it by north-south–trending thrust faults (Figs. 1 and 2).

GEOLOGY

The Visean to Late Carboniferous succession sampled is within the Tamworth Belt of the southern New England orogen (Fig. 1). The belt is situated between the Central Complex of the orogen to the east and the Hunter-Mooki thrust system to the west. Carboniferous rocks of the belt were deposited within a forearc basin adjacent to an andesitic, dacitic, and rhyolitic continental arc. Tournaisian marine shelf sediments were overlain in the Visean by volcanogenically derived fluvial sedimentary rocks containing lavas and tuffs, and minor intercalations of fossiliferous marine shelf deposits. Uplift toward the end of the Visean caused erosion and a gap in the record. This late Visean to early Namurian hiatus had a duration of about 20 m.y. in the Werrie syncline and probably less than 10 m.y. in the Rocky Creek syncline and adjacent blocks. The hiatus was shorter in the region southeast of the Liverpool Range and one continuous sequence is known from the Stroud-Gloucester syncline (Roberts et al., 1995a). In the region sampled, this hiatus effectively divides the Carboniferous succession into lower and upper sequences. Sedimentary rocks deposited following the hiatus consist of continental glaciogene, volcanogenic, and fluvial successions of Namurian, Westphalian, and possibly early Stephanian age. This hiatus is not important for the present study because, as described herein, the base of the Kiaman is above it within the Rocky Creek syncline.

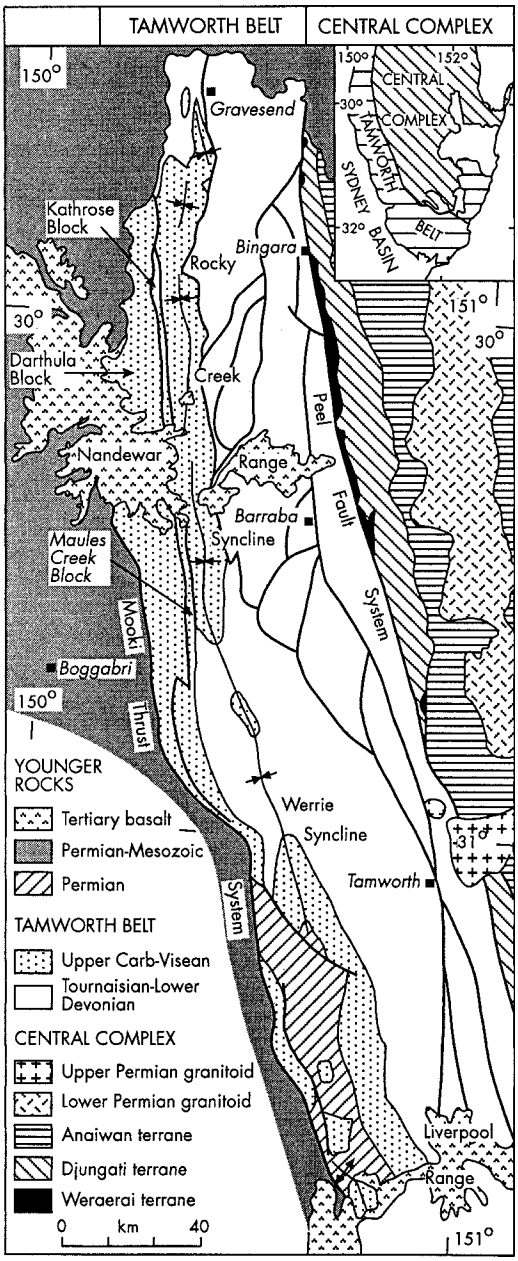


Figure 1. Regional geological map of the northwestern part of the southern New England orogen indicating the relationship between the Tamworth Belt and the Central Complex of New England. The regions sampled include the Rocky Creek and Werrie synclines and Kathrose, Maules Creek, and Darthula blocks at the western margin of the Tamworth Belt. Adapted from Pogson (1972). Terranes of the Central Complex are those of Flood and Aitchison (1988). Tertiary basalt has been omitted from northern parts of the Central Complex.

Although the Rocky Creek and Werrie synclines (Figs. 2 and 3) share the same fold axis, the stratigraphy of the Visean to Late Car-

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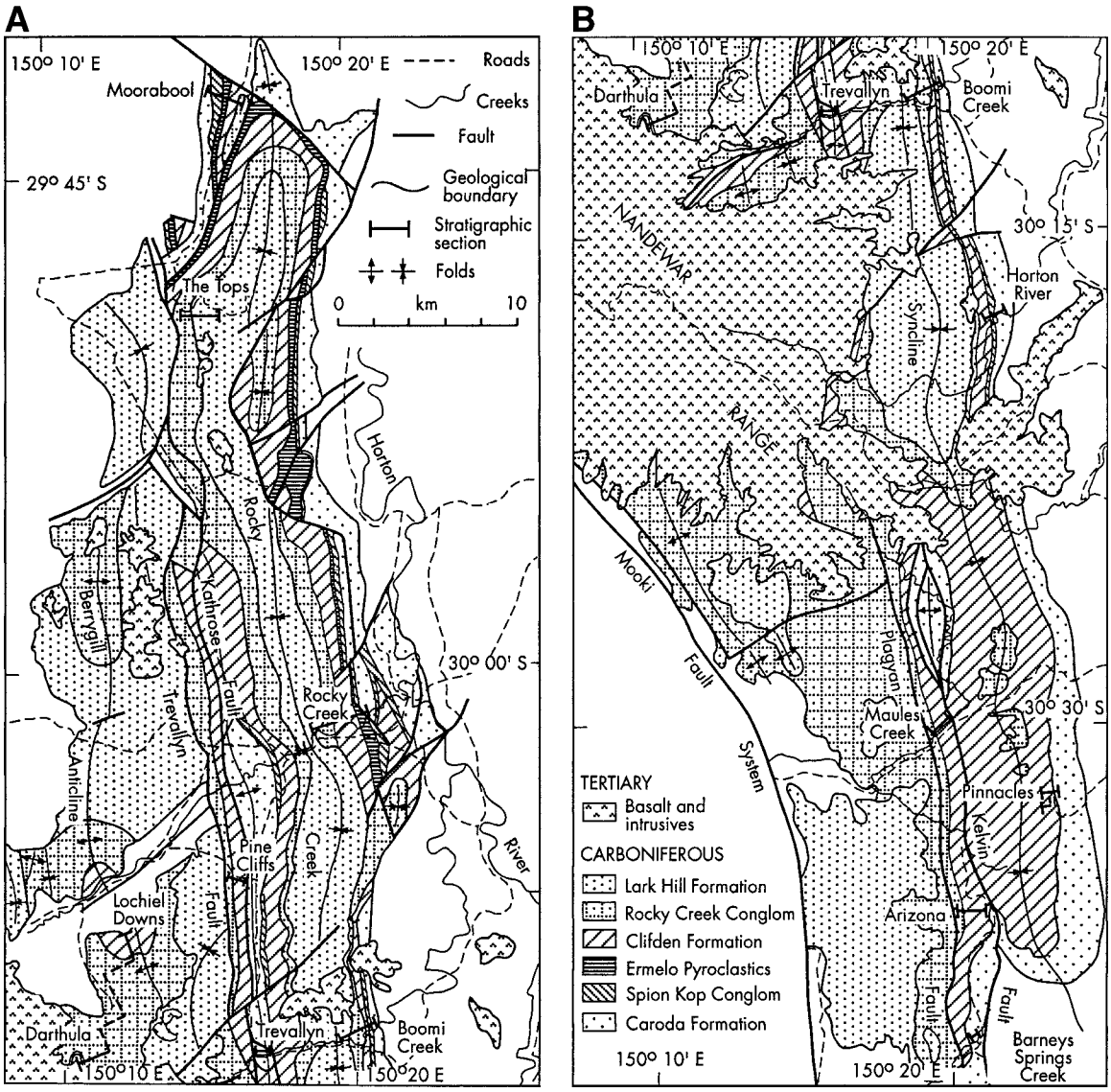


Figure 2. Geological map of Visean to Late Carboniferous formations of the Rocky Creek syncline and adjacent areas indicating the location of the measured stratigraphic sections sampled. Modified from the Gravesend, Horton, and Boggabri 1:100 000 sheets (Brown, 1987a, 1987b; Stroud, 1990).

boniferous succession changes suddenly at the southern closure of the Rocky Creek syncline. To the north, throughout the Rocky Creek syncline and the Kathrose–Maules Creek and Darthula blocks, the upper succession is divided into five formations. By contrast, to the south, throughout the Werrie syncline as far as the Liverpool Range, where there are distinctive early to middle Visean sedimentary and volcanic rocks, the hiatus is longer, extending from the late Visean to Westphalian, and the upper sequence can be mapped only as a single formation.

Rocky Creek Syncline Region

Lower sequence rocks of the Rocky Creek syncline are represented by the Lower Carboniferous Luton (oldest), Namoi, and Caroda Formations. Only the latter contains volcanic rocks and has been sampled.

The Caroda Formation is the oldest unit sampled and transitionally overlies the Tournaisian Namoi Formation. It contains fossiliferous sandstone, oolitic limestone, and heavy mineral sandstone lenses overlain by coarse fluvial lithic sandstone, lensoidal pebble conglomerate, minor carbonaceous shale, and volcanic rocks (Fig. 4). The marine rocks are middle Visean (Jenkins et al., 1993; Roberts et al., 1993). In the northern Rocky Creek syncline, the Caroda Formation contains dacitic to rhyodacitic ignimbrites (Boomi Rhyodacite and Yuendoo Rhyolite) and a rare flow of andesite (Kooringal Andesite), whereas in the south the unit is dominated by the thick Barneys Springs Andesite. The Caroda Formation has been sampled for paleomagnetic study at four sites at three localities (Fig. 4).

The Spion Kop Conglomerate is the lowest unit of the upper sequence. It disconformably overlies the Caroda Formation, and consists of fluvial to glaciogene conglomerate and diamictite (White, 1968). Because glaciation in the southern New England orogen did not commence until the beginning of the Namurian (Roberts et al., 1995a), the

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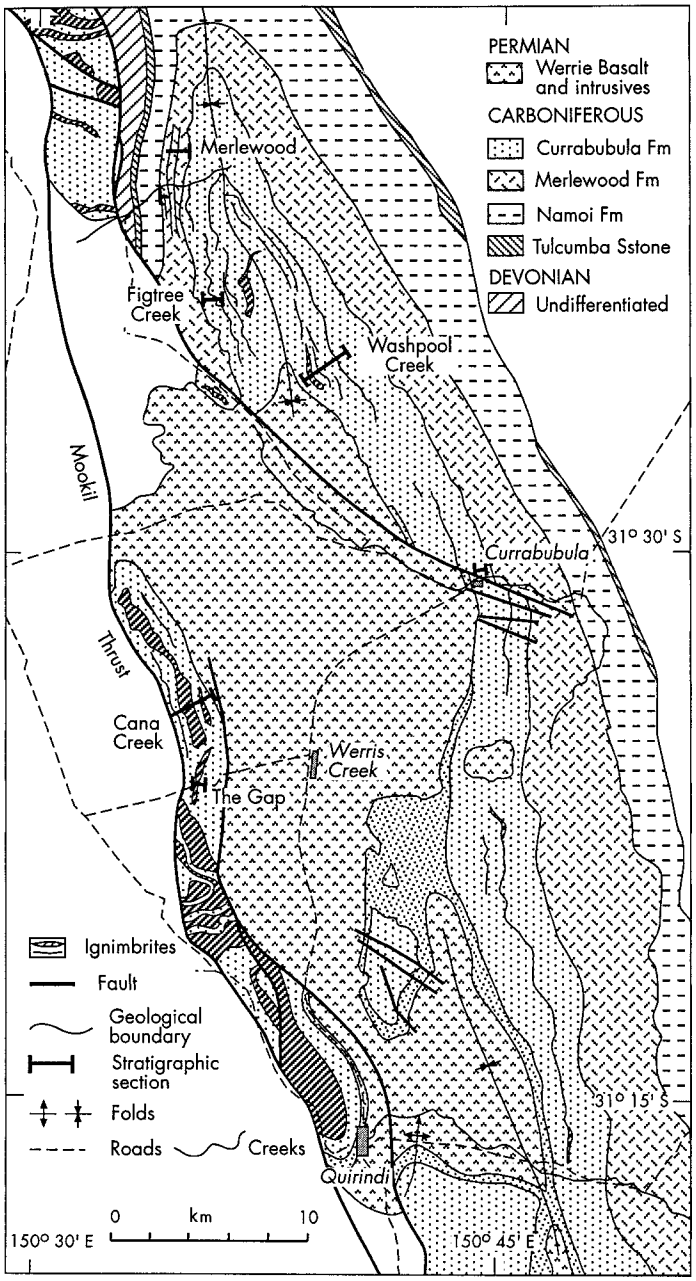


Figure 3. Geological map of the Carboniferous and Permian units in the Werrie syncline indicating the location of the measured stratigraphic sections sampled. Modified from the Tamworth 1: 250 000 sheet (Offenberg, 1967) using the work of McPhie (1983). Mafic to intermediate Permian intrusive rocks have not been differentiated from the Werrie Basalt.

Spion Kop Conglomerate is interpreted as early Namurian in age. The conglomerate is as much as 550 m thick, widespread throughout most of the Rocky Creek syncline, and is frequently lensoidal. We found no rocks in this sequence suitable for paleomagnetic work or dating.

In northern parts of the Rocky Creek syncline the Spion Kop Conglomerate is overlain by the Ermelo Pyroclastics, a partly resedimented crystal-rich pyroclastic as much as 300 m thick that is interpreted as a single erupted unit. This unit has been dated by the hornblende K-Ar method, giving an age of 319 ± 5 Ma (J. Roberts, unpublished data). Zircons extracted from the Ermelo Pyroclastics sampled at site 40 (site numbers are located in Table 1; Figs. 4–6) have very simple U-Pb isotopic compositions (Table DR11). The raw data are plotted in a concordia diagram in Figure 7A and cluster close to the concordia curve, indicating that the very small contents of common Pb are a trivial influence on calculated ages. The $^{206}Pb/^{238}U$ ratios of the individual analyzed zircon zones cohere tightly, indicating homogeneous ages. This coherence is tested in Figure 7B, where the individual $^{206}Pb/^{238}U$ ages are ranged against an x-axis scaled as probability. The close adherence of the data to the theoretical normal distribution is substantiated by an acceptable mean square of weighted deviates (MSWD). There is no evidence that inherited ages or alteration have affected the isotopic systems. The weighted mean age of 41 analyses is 321.3 ± 3.2 Ma (2σ), which is interpreted as the emplacement age of the Ermelo Pyroclastics.

The Clifden Formation, which extends throughout the Rocky Creek region, overlies either the Spion Kop Conglomerate or, where present, the Ermelo Pyroclastics. It contains as much as 700 m of fluvial feldspatholithic sandstone, plant-bearing siltstone, conglomerate, lacustrine ash-rich cherty siltstone, mudstone, and ignimbrites and other volcanic units. Four volcanic units within the Clifden Formation have been dated. The lowermost regionally widespread unit is the Wanganui Andesite (Nott, 1968), a discontinuous black to green pyroxene andesite confined to the southern half of the syncline and the Maules Creek block. Zircons from site 113 have a simple range of compositions. The raw data cluster close to concordia, indicating very low contents of common Pb (Fig. 8A). Homogeneous $^{206}Pb/^{238}U$ ratios indicate a dominant crystallization age. Homogeneity is confirmed by the main population of ages adhering closely to that expected for a normal distribution (Fig. 8B). A single zircon has a discrepant old age and is identified as a ca. 340 Ma xenocryst; one grain with a young apparent age of ca. 295 Ma also departs from the norm and is interpreted to have leaked radiogenic Pb. The remaining 41 analyses group with acceptable MSWD and indicate a mean crystallization age of 319.2 ± 2.8 Ma (2σ), which is interpreted as the eruption age of the Wanganui Andesite.

The Wanganui Andesite is overlain by the Eastons Arm Rhyolite (Nott, 1968), a pale gray-green moderately welded ignimbrite containing sparse crystals of quartz and feldspar in a fine-grained groundmass. Zircon data from site 111 have slightly higher proportions of common Pb, but correction along the vector shown (equivalent to crustal Pb of the same age as the zircons) produces appropriate correction onto concordia (Fig. 8C). The range of radiogenic $^{206}Pb/^{238}U$ ages is tested in a probability diagram in Figure 8D, which highlights a single ca. 360 Ma xenocryst and one grain with an apparent age lower than the norm, indicating Pb loss. The remaining 41 zircon compositions conform appropriately to the expected normal distribution with acceptable MSWD and a weighted mean crystallization age of 317.8 ± 2.8 Ma (2σ), interpreted as the emplacement age of the Eastons Arm Rhyolite.

The Wanganui Andesite–Eastons Arm Rhyolite samples were dated together on the same SHRIMP mount. Consequently, their relative ages are established independently from the uncertainties of calibration to the SL13 standard; only their absolute ages depend on reference to the standard. The Eastons Arm Rhyolite is slightly younger than the Wanganui Andesite, as superposition requires.

[1]GSA Data Repository item 200092, U-Pb isotopic data tables, is available on the Web at http://www.geosociety.org/pubs/ft2000.htm. Requests may also be sent to Documents Secretary, GSA, P.O. Box 9140, Boulder, CO 80301; e-mail: editing@geosociety.org.

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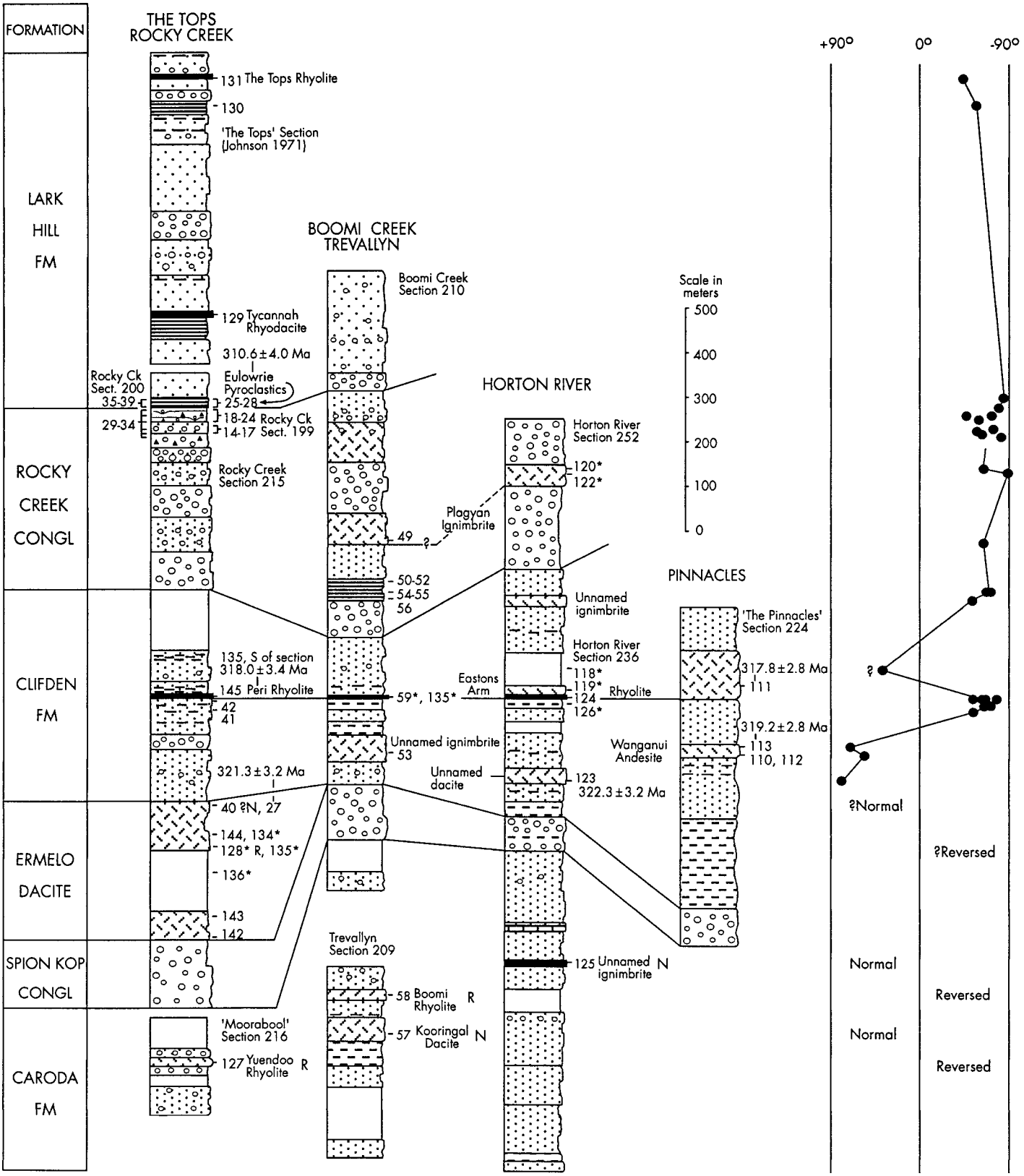


Figure 4. Measured sections of Visean to Late Carboniferous strata within the Rocky Creek syncline indicating stratigraphic positions of paleomagnetic sampling sites. White parts of columns indicate no outcrop. Locations of section lines are given in Figure 2. Section numbers are those of J. Roberts. Virtual geomagnetic pole (VGP) latitudes are given on right; positive (negative) indicates normal (reversed) polarity.

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TABLE 1. PALEOMAGNETIC DIRECTIONS BY SITES

No.	Lat (°S)	Long (°E)	dip/DDA	*N*	D_B, I_B	pol	*k*	α_{95}	Formation	Lithology
1	30.144	150.1992	03/190	6	221,73	R	28	13	CLF	Red ignimbrite
2	30.143	150.1992	03/175	3	197,65	R	11	40	CLF	Downs Rhyodacite
3	30.139	150.2003	03/165	3	152,87	R	12	37	CLF	Ash-rich varvoid
4	30.139	150.2003	04/180	2	157,76	R	N.D.	N.D.	CLF	Ash-rich varvoid
5	30.139	150.2003	04/240	1	206,75	R	N.D.	N.D.	CLF	Ash-rich varvoid
6	30.139	150.2003	04/210	1	214,78	R	N.D.	N.D.	CLF	Ash-rich varvoid
7	30.138	150.1972	06/215	3	slump test		N.D.	N.D.	CLF	Ash-rich varvoid
8	30.136	150.1967	06/210	7	209,73	R	284	4	CLF	Red ignimbrite
9	30.195	150.1830	17/275	5	208,74	R	736	3	RCC	Darthula Rhyodacite
10	30.194	150.1830	22/270	5	225,52	R	1125	2	RCC	Darthula Rhyodacite
11	30.189	150.1844	24/250	5	215,64	R	250	5	RCC	Red ignimbrite
12	30.191	150.1822	20/260	6	208,70	R	388	3	RCC	Darthula Rhyodacite
13	30.161	150.1892	16/270	5	217,79	R	68	9	RCC	Mount Hook Ignimbrite
14	30.034	150.3136	22/275	3	257,80	R	26	25	RCC	Fine gray-green ss.
15	30.034	150.3131	20/275	2	267,77	R	N.D.	N.D.	RCC	Fine gray-green sandstone
16	30.033	150.3131	20/275	3	198,76	R	126	11	RCC	Fine gray-green sandstone
17	30.033	150.3122	15/225	3	172,69	R	8	47	RCC	Coarse gray-green sandstone
18	30.034	150.3122	12/215	2	106,77	R	N.D.	N.D.	RCC	Brown-green sandstone
19	30.034	150.3122	12/215	4	064,85	R	30	17	RCC	Green siltstone
20	30.034	150.3122	12/215	14	congl. test		2	N.D.	RCC	Igneous clasts
21	30.034	150.3117	21/240	2	224,49	R	N.D.	N.D.	RCC	Fine siltstone
22	30.034	150.3117	18/235	3	187,85	R	100	12	RCC	Varve
23	30.033	150.3111	15/235	3	096,79	R	21	27	RCC	Varve
24	30.033	150.3111	15/235	2	incoherent		N.D.	N.D.		Banded coarse sandstone
25	30.033	150.3106	14/233	3	179,80	R	282	7	RCC	Pink ash bed
26	30.033	150.3100	14/233	2	217,83	R	N.D.	N.D.	RCC	Pink tuffaceous siltstone
27	30.034	150.3094	14/233	2	141,87	R	N.D.	N.D.	RCC	Pink tuffaceous siltstone
28	30.034	150.3088	10/233	3	178,77	R	318	7	RCC	Pink tuffaceous siltstone
29	30.043	150.3036	19/060	3	200,75	R	50	18	RCC	Gray diamictite
30	30.043	150.3036	19/060	3	174,64	R	19	30	RCC	Gray diamictite
31	30.043	150.3036	19/060	1	157,67	R	N.D.	N.D.	RCC	Gray diamictite
32	30.034	150.3053	25/065	4	112,56	R	52	13	RCC	Gray diamictite
33	30.034	150.3053	25/065	3	168,61	R	38	20	RCC	Gray-green varve
34	30.042	150.3047	25/065	3	166,85	R	26	25	RCC	Gray varve
35	30.042	150.3048	16/065	3	189,82	R	405	6	LaF	Silty ash
36	30.042	150.3061	16/065	3	208,79	R	289	7	LaF	Cherty ash
37	30.042	150.3066	10/070	3	205,74	R	252	8	LaF	Cherty ash
38	30.042	150.3078	10/070	3	192,82	R	135	11	LaF	Ash-rich sandstone
39	30.050	150.3103	0/0	3	190,78	R	131	11	LaF	Ash-rich sandstone
40	30.031	150.3386	42/242	6	incoherent		N.D.	N.D.		Ermelo Dacite
41	30.033	150.3356	12/230	3	094,76	R	25	25	CLF	Gray lithic sandstone
42	30.032	150.3339	30/230	3	205,58	R	132	11	CLF	Gray lithic sandstone
43	30.107	150.2769	55/290	3	071,64	R	94	13	CLF	Ash-rich siltstone
44	30.106	150.2667	58/255	4	162,62	R	312	5	CLF	Pine Cliffs Rhyodacite
45	30.107	150.2650	70/240	2	incoherent		N.D.	N.D.	CLF	Pink unwelded ignimbrite
46	30.107	150.2650	60/260	3	149,69	R	21	28	CLF	Coarse brown ignimbrite
47	30.107	150.265	67/260	2	201,80	R	30	48	CLF	Unwelded ignimbrite
48	30.103	150.2708	85/260	5	196,55	R	171	6	CLF	Bexley Rhyodacite
49	30.187	150.3339	47/260	5	223,69	R	6	35	RCC	Dacite
50	30.187	150.3350	47/260	2	117,77	R	N.D.	N.D.	RCC	Ash-rich siltstone
51	30.187	150.3350	47/260	3	223,65	R	51	18	RCC	Ash-rich siltstone
52	30.187	150.3350	47/260	2	111,77	R	N.D.	N.D.	RCC	Ash-rich siltstone
53	30.184	150.3392	65/260		incoherent				CLF	Coarse unwelded ignimbrite
54	30.203	150.3361	70/260	2	238,77	R	N.D.	N.D.	RCC	Gray siltstone
55	30.192	150.3369	55/260	3	225,82	R	115	12	RCC	Gray siltstone
56	30.193	150.3375	65/260	2	103,72	R	N.D.	N.D.	RCC/CLF	Fine, lithic sandstone
57	30.195	150.2786	20/245	5	56,–63	N	7	31	Car	Kooringal Andesite
58	30.196	150.2772	20/245	9	104,69	R	114	5	Car	Boomi Rhyodacite
59	30.192	150.2903	12/090	8	204,68	R	87	6	CLF	Peri Rhyodacite
60	31.120	150.5858	30/090	5	146,83	R	129	7	CuF	Red ignimbrite
61	31.119	150.5869	24/110	4	190,67	R	138	8	CuF	Fine red ash
62	31.119	150.5869	12/095	4	209,68	R	37	15	CuF	Fine red ash
63	31.119	150.5881	18/095	3	247,83	R	10	42	CuF	Red ignimbrite
64	31.120	150.5903	25/090	2	242,64	R	N.D.	N.D.	CuF	Red-purple tuff
65	31.118	150.5914	25/090	4	150,80	R	290	5	CuF	Red-purple ignimb.
66	31.118	150.5914	25/090	5	168,83	R	405	4	CuF	Red-purple ignimb.
68	31.118	150.5933	25/090	3	226,73	R	75	14	CuF	Red-purple ashy sls.
69	31.115	150.5956	25/090	3	270,83	R	336	7	CuF	Cana Creek Tuff
70	31.116	150.5986	24/115	4	212,68	R	471	4	CuF	Red fine ash-rich sls.
71	31.085	150.4581	42/65	3	314,–21	N	273	8	MF	Andesite
72	31.583	150.5722	40/88	2	184,35	R	N.D.	N.D.	MF	Andesite
73	31.166	150.6442	28/260	6	280,67	R	255	4	CuF	Piallaway Ignimbrite
74	31.166	150.6467	42/245	4	226,76	R	43	14	CuF	Red ash-rich tuff
75	31.166	150.6483	36/235	3	212,66	R	530	5	CuF	Red siltstone varve
76	31.166	150.2336	32/240	5	232,58	R	71	9	CuF	Taggarts Mt. Ignimbrite
77	31.165	150.6519	40/240	3	scattered	R	N.D.	N.D.	CuF	Red ignimbrite/ash
78	31.165	150.6531	15/250	3	071,68	R	66	15	CuF	Purple ash
79	31.164	150.6542	42/250	5	256,70	R	104	8	CuF	Pink ignimbrite/ash

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TABLE 1. (*Continued*)

No.	Lat (°S)	Long (°E)	dip/DDA	*N*	D_B, I_B	pol	*k*	α_{95}	Formation	Lithology
80	31.166	150.6572	42/250	2	incoherent				CuF	Purple-orange ignimbrite
81	31.166	150.6556	37/248	4	256,80	R	128	8	CuF	Green ash
82	31.166	150.6556	32/250	3	189,88	R	238	8	CuF	Buff tuff
83	31.167	150.6603	32/250	3	231,80	R	203	9	CuF	Cana Creek Tuff
84	31.164	150.6625	32/250	2	124,69	R	N.D.	N.D.	CuF	Diamictite
85	31.164	150.6636	33/240	2	064,65	R	N.D.	N.D.	CuF	Fine red ash
86	31.164	150.6678	32/245	2	023,62	R	N.D.	N.D.	CuF	Gray tuffaceous sandstone
87	31.310	150.5733	35/063	8	254,71	R	160	4	CuF	Cana Creek Tuff
88	31.311	150.5756	48/090	1	193,56	R	N.D.	N.D.	CuF	Red varve
89	31.306	150.5756	48/090	5	218,67	R	425	4	CuF	red varve
90	31.473	150.5761	30/060	10	congl. test	N.D.	1	N.D.	CuF	Igneous clasts
91	31.473	150.5664	*30/060	3	congl. test	N.D.	N.D.	N.D.	CuF	Igneous clasts
92	31.305	150.5792	20/080	5	251,68	R	631	3	CuF	Taggarts Mt. Ignimbrite
93	31.305	150.5797	35/045	4	281,60	R	9108	1	CuF	Taggarts Mt. Ignimbrite
94	31.304	150.5803	42/065	2	204,70	R	N.D.	N.D.	CuF	Purple varves
95	31.304	150.5803	55/065	4	181,72	R	64	12	CuF	Purple varves
96	31.304	150.5814	58/070	6	114,76	R	348	4	CuF	Red ignimbrite
97	31.303	150.5828	52/065	6	250,70	R	727	3	CuF	Purple-brown ignimbrite
99	31.358	150.5947	32/112	7	258,72	R	15	16	CuF	Ignimbrite
100	31.358	150.5947	28/110	5	229,68	R	19	18	CuF	Varved diamictite
101	31.358	150.5947	35/090	1	poor	R	N.D.	N.D.	CuF	Varved diamictite
102	31.358	150.5956	30/110	5	056,83	R	18	19	CuF	Fine diamictite
103	31.358	150.5656	36/105	5	173,62	R	4	45	CuF	Red-purple diamictite
104	30.662	150.3495	43/240	5	262,–69	N	9	28	Ca	Barneys Sp. Andesite
105	30.662	150.3481	35/240	5	incoherent				Ca	Barneys Sp. Andesite
106	30.663	150.3478	43/245	5	024,–34	N	11	25	Ca	Barneys Sp. Andesite
107	30.662	150.3498	43/245	5	358,–65	N	17	19	Ca	Red tuff
108	30.089	150.3692	63/235	5	023,–40	N	21	17	ClF	Wanganui Andesite
109	30.589	150.3505	55/250	5	152,69	R	428	4	ClF	Eastons Arm Rhyodacite
110	30.527	150.3922	32/275	5	incoherent		N.D.	N.D.	ClF	Red tuff
111	30.527	150.3904	42/255	4	223,53	R	522	4	ClF	Eastons Arm Rhyodacite
112	30.531	150.3930	22/285	7	011,–50	N	13	17	ClF	Wanganni Andesite
113	30.531	150.3918	24/285	6	346,–64	N	167	5	ClF	Wanganni Andesite
114	30.505	150.3317	52/235	5	incoherent				RCC	Sandstone
115	30.505	150.3317	55/250	3	081,76	R	26	25	ClF	Gray ignimbrite
116	30.502	150.3346	53/250	3	098,86	R	122	11	ClF	Red-green ignimbritic
117	30.501	150.3353	60/255	1	190,83	R	N.D.	N.D.	ClF	Appleogue Dacite
118	30.471	150.3923	30/298	5	081,–56	N	30	14	ClF	Red ignimbrite
119	30.449	150.3574	13/100	5	scattered	R	N.D.	N.D.	ClF	Dacite
120	30.463	150.3733	00/000	5	130,72	R	152	6	RCC	Plagyan Rhyolite
121	30.394	150.3697	32/255	4	193,74	R	118	9	ClF	Peri Rhyolite
122	30.381	150.3453	12/030	6	184,77	R	3173	1	RCC	Plagyan Phyolite
123	30.295	150.3708	45/275	5	353,–70	N	38	13	ClF	Dacite
124	30.300	150.3698	35/275	5	170,70	R	171	16	ClF	Peri Rhyolite
125	30.295	150.3762	52/270	6	68,–55	N	9	23	Ca	Red ignimbrite
126	30.260	150.3605	57/256	5	193,65	R	42	12	ClF	Gray siltstone
127	29.720	150.2615	32/80	6	357,26	R	3	52	Ca	Yuendoo Rhyolite
128	29.713	150.2695	05/090	5	313,25	R	108	7	Ed	Ermelo Dacite
129	29.822	150.2447	28/080	5	incoherent		N.D.	N.D.	LaF	Tycannah Dacite
130	29.817	150.2533	30/080	5	118,71	R	28	15	LaF	Gray ash
131	29.823	150.2531	15/075	4	087,69	R	11	29	LaF	The Tops Dacite
132	30.057	150.3667	17/215	6	113,79	R	113	6	ClF	Fine gray ash
133	30.057	150.3667	14/235	5	incoherent	N.D.	N.D.	N.D.	ClF	Red ignimbrite
134	30.115	150.2860	22/095	5	182,51	R	443	4	ClF	Glen Idle Ignimbrite
135	30.113	150.2878	24/030	5	132,84	R	78	9	ClF	Peri Rhyolite
136	29.904	150.3016	18/273	5	overprinted	N.D.	N.D.	N.D.	Ed	Ermelo Dacite
137	29.904	150.3022	18/290	5	overprinted	N.D.	N.D.	N.D.	Ed	Ermelo Dacite
138	29.905	150.3048	18/250	5	overprinted	N.D.	N.D.	N.D.	Ed	Ermelo Dacite
139	30.178	150.2691	40/275	5	252,87	R	320	4	ClF	Andesite
140	30.177	150.2665	40/275	5	141,84	R	587	3	ClF	Andesite
141	30.177	150.2665	35/260	5	scattered	R	N.D.	N.D.	ClF	Ash bed
142	30.026	150.3424	70/240	5	overprinted	N.D.	N.D.	N.D.	Ed	Ermelo Dacite
143	30.025	150.3414	70/240	5	overprinted	N.D.	N.D.	N.D.	Ed	Ermelo Dacite
144	30.029	150.3402	42/252	5	overprinted	N.D.	N.D.	N.D.	Ed	Ermelo Dacite
145	30.031	150.3369	19/225	5	166,60	R	248	5	ClF	Peri Rhyolite

Note: Data are from the Ermelo Dacite (Ed), Caroda Formation (Ca), Clifden Formation (ClF), Rocky Creek Conglomerate (RCC), and Lark Hill (LaF) in the Rocky Creek area, and the Merlewood (MF) and Currabubula (CuF) Formations in the Werrie Syncline. Collecting site numbers in the left-hand column. Bedding attitude specified by dip and DDA (down-dip azimuth). D_B,I_B declination and inclination relative to bedding. Pol (polarity), R (reversed), N (normal). *k* and α_{95} (Fisher's precision and half angle of cone of confidence); $P = 0.05$. N.D.—not determined.

*Combined with 90.

Field, petrographic, and geochemical studies (Xiang Wang, unpublished data) indicate that the Eastons Arm Rhyolite is the same unit as the Peri Rhyolite, described by McKelvey and White (1964) from the Kathrose block to the northwest and mapped throughout most of the northern part of the Rocky Creek syncline (Brown, 1987b; Stroud, 1990). Zircons from the Peri Rhyolite (site 135) have very high contents of common Pb, ranging to 50%; many of the uncorrected compositions are off scale in the concordia diagram (Fig. 9A). High to

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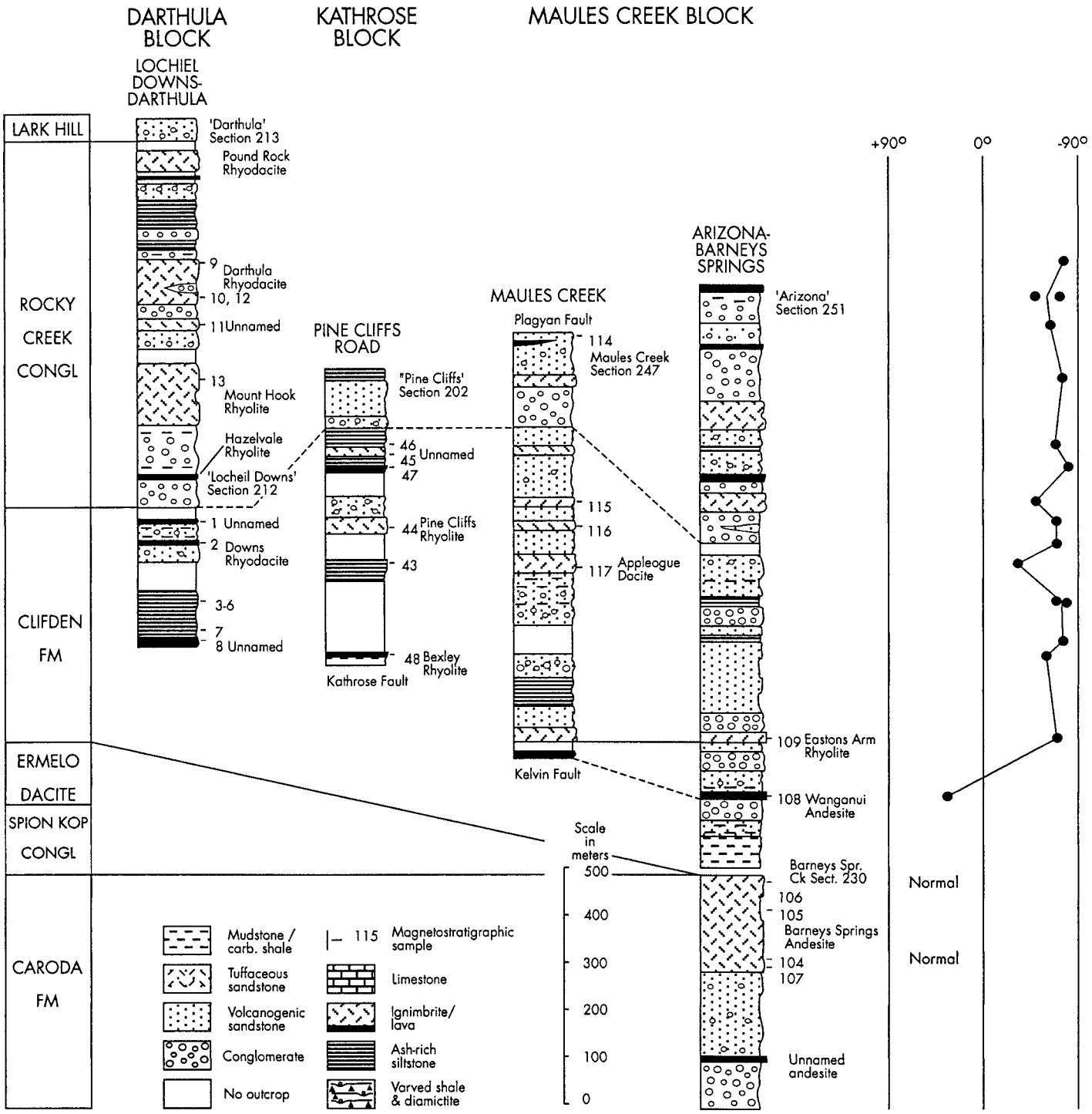


Figure 5. Measured sections of Visean to Late Carboniferous strata within the Darthula, Kathrose, and Maules Creek blocks indicating stratigraphic positions of paleomagnetic sampling sites. See Figure 4 caption.

extreme common Pb correlates with elevated contents of U and Th. This contrasts with zircons in the Eastons Arm Rhyolite sample, which, although having a similar range of U and Th contents, have trivial common Pb abundances, typically below 1%. Correction for common Pb (Fig. 9A) results in the complex age spectrum displayed in the probability diagram (Fig. 9B). There are two old $^{206}Pb/^{238}U$ ages identifying ca. 350 Ma xenocrysts. The remainder partly conforms to the expected normal distribution, but also has a distinct tail of young ages in which it is difficult to establish a confident statistical cutoff. This tail comprises 10 grains with very high contents of U and high common Pb, interpreted as having undergone open-system exchange of radiogenic and common Pb. The remaining 27 analyses form a ho-

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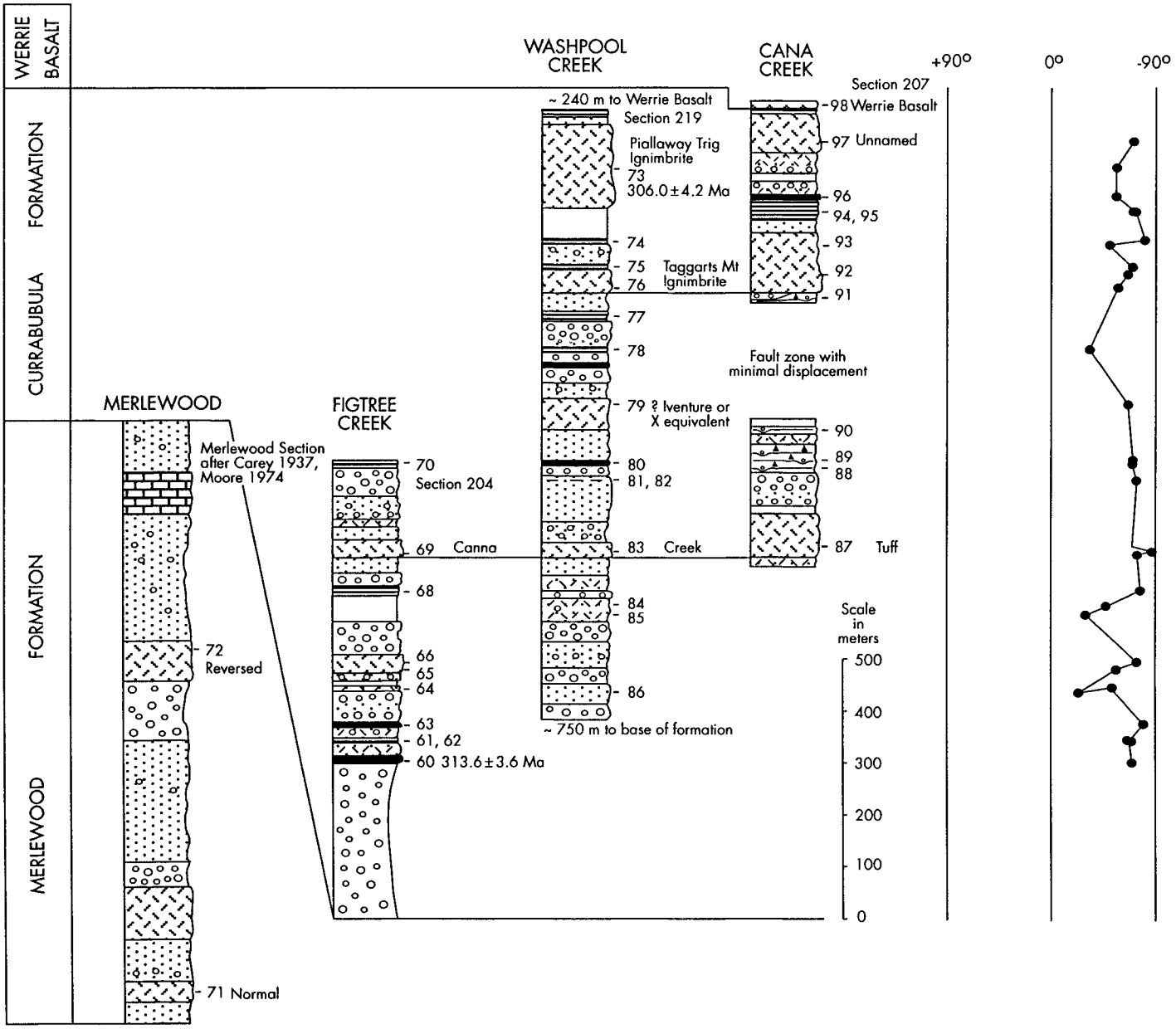


Figure 6. Measured sections of Visean to Late Carboniferous strata within the Werrie syncline indicating stratigraphic positions of paleomagnetic sampling sites. See Figure 4 caption.

mogeneous grouping with acceptable MSWD and a mean age of 318.0 ± 3.4 Ma (2σ). Whereas this age is interpretation dependent and not as well established as that of the Eastons Arm Rhyolite, the similarity in ages supports field, petrological, and geochemical evidence that they are products of the same eruptive event.

Beneath the Peri Rhyolite there are several less regionally extensive unnamed volcanic units at about the same stratigraphic level as the Wanganui Andesite. These include an unwelded red ignimbrite (site 53) in the Boomi Creek section; the Glen Idle Rhyolite (site 134; Hocking, 1973) on the western limb of the Rocky Creek syncline north of Trevallyn; and an unnamed hornblende dacite in the Horton River section (site 123). Zircons from the latter have simple U-Pb isotopic compositions. The concordia diagram (Fig. 9C) illustrates their very low contents of common Pb, and their ages adhere closely to the expected normal distribution (Fig. 9D). The weighted mean age of the zircon population is 322.3 ± 3.2 Ma (2σ), distinctly older than the 319.2 ± 2.8 Ma Wanganui Andesite.

Within a complex zone of faulting, 6 km north of the Pinnacles (Fig. 2B), an unnamed red ignimbrite (site 118) appears to overlie the Eastons Arm Rhyolite. Exposure of the ignimbrite is restricted to a single small exposure in a roadside drain. No faults are exposed, but the fracture pattern on the air photographs and the locations of displaced blocks of the Eastons Arm Rhyolite are such that the most likely stratigraphic location for the unnamed red ignimbrite is about 50 m above the rhyolite.

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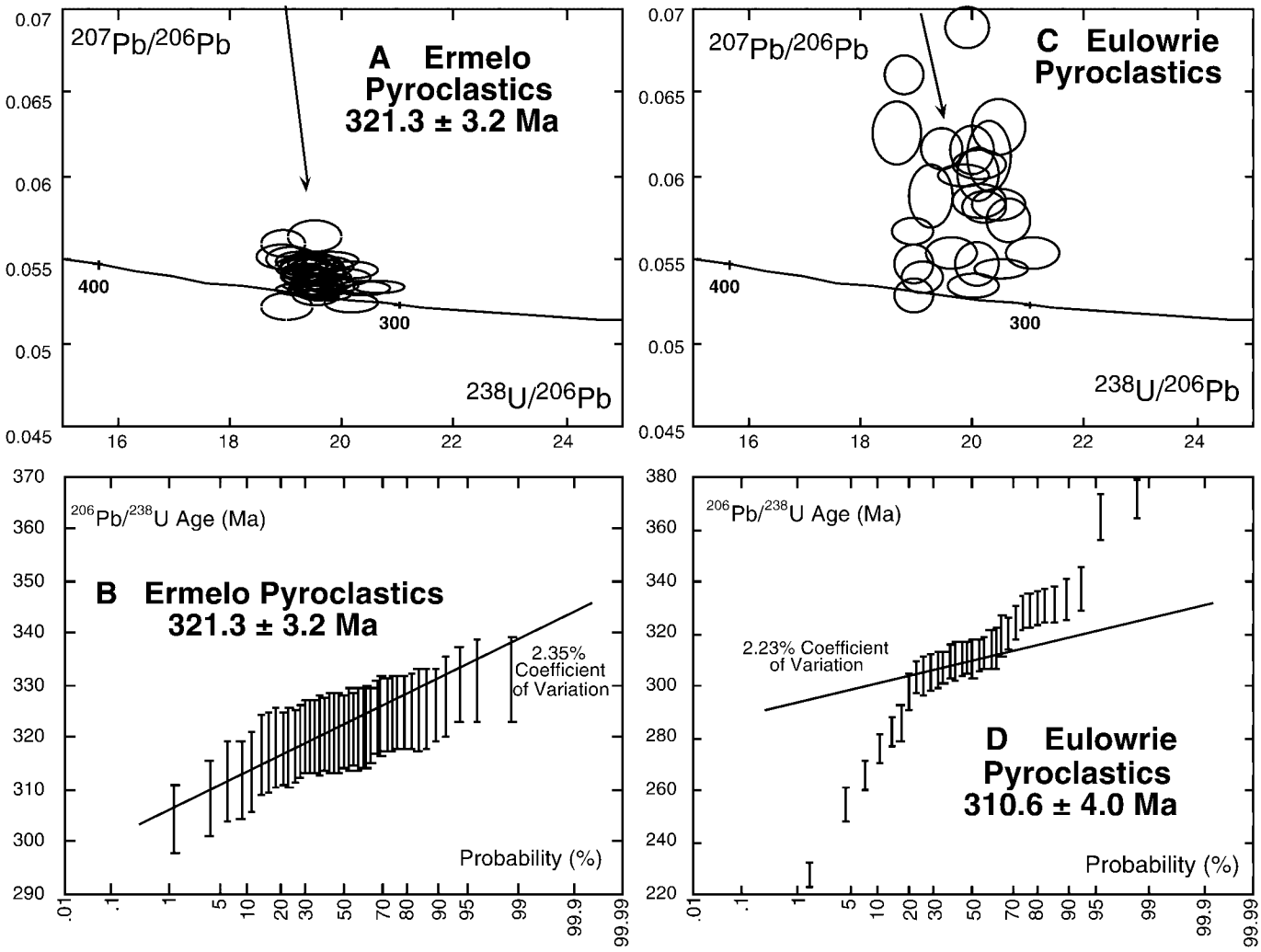


Figure 7. Concordia and probability diagrams of zircon U-Pb isotopic compositions in the Ermelo Pyroclastics and Eulowrie Pyroclastics. The homogeneous compositions in the Ermelo sample define an age of 321.3 ± 3.2 Ma. The complex array of ages in the Eulowrie sample indicates mixing of magmatic zircons with xenocrystic components, as well as leakage of isotopic systems in some zircon grains.

In the Maules Creek block southwest of the Rocky Creek syncline (Fig. 2B), the Wanganui Andesite and Eastons Arm Rhyolite are succeeded in some sections by the Appleogue Dacite, a dark gray to black hornblende dacite, and multicolored unwelded ignimbrites (Fig. 5). The structurally equivalent Kathrose block situated north of the Nandewar Range contains as many as four ignimbrites and numerous multicolored ash-rich siltstone units. The Peri Rhyolite, which crops out north of Rocky Creek, cannot, because of lack of exposure, be linked with sections to the south, such as that at Pine Cliffs (Fig. 5).

The westernmost Darthula block contains only the uppermost part of the Clifden Formation. There are three unwelded red ignimbrites, including the Downs Rhyodacite (Hocking, 1973), and thick, gray, ash-rich lacustrine siltstones.

The Rocky Creek Conglomerate conformably overlies the Clifden Formation and is as much as 800 m thick (McKelvey and McPhie, 1985). It contains volcanic and fluvial cobble conglomerate, minor sandstone, tillite, and glaciolacustrine varved shales. Glaciogene rocks are largely restricted to the upper part of the formation within the axial region of the Rocky Creek syncline. McKelvey and McPhie (1985) interpreted the Rocky Creek Conglomerate to be piedmont fan deposits.

The major volcanic unit is the Plagyan Ignimbrite, a buff, purple, or gray ignimbrite with welded rhyolitic and unwelded dacitic cooling units. It is best developed on the southern flank of the Nandewar Range within the southern extension of the Darthula block, where it reaches a maximum thickness of 300 m. It has also been mapped throughout the Rocky Creek syncline as far north as Boomi Creek, but it has not yet been identified within the Darthula block north of the Nandewar Range (Fig. 5). As many as six ignimbrites, mostly unwelded to moderately welded red to brown units, and thick resedimented ash-rich siltstone, are present within the Rocky Creek Conglomerate in the Darthula block. The U/Pb ages of units above and below the Rocky Creek Conglomerate suggest that the formation ranges in age from early to middle Westphalian. Irving's (1966) samples from the Rocky Creek syncline were taken from glaciogene rocks of the uppermost Rocky Creek Conglomerate and resedimented ash within the overlying Eulowrie Pyroclastics of the Lark Hill Formation.

The Lark Hill Formation is at the top of the upper sequence and

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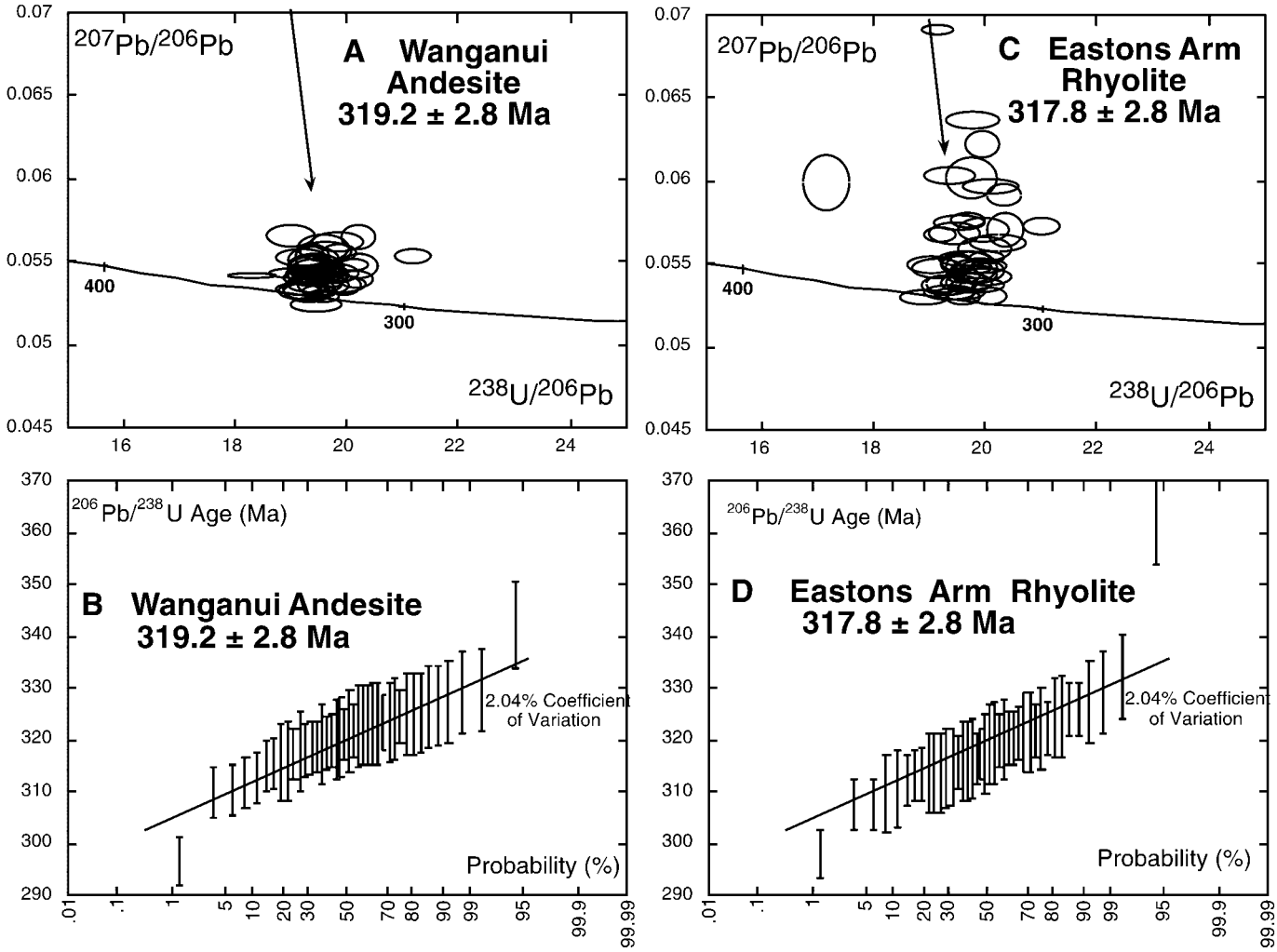


Figure 8. Concordia and probability diagrams illustrating zircon compositions in the Wanganui Andesite and Eastons Arm Rhyolite. Both have simple arrays of zircon ages defining confident magmatic crystallization ages.

comfortably overlies the Rocky Creek Conglomerate. It is composed of coarse volcanogenic lithic sandstone, pebble conglomerate, pebbly sandstone, and siltstone deposited in upward-fining cycles, and disrupted by pyroclastic ash falls and ignimbrites. Pyroclastic units are developed mainly within the northern part of the Rocky Creek syncline. In the region between Rocky Creek and Trevallyn, the base of the Lark Hill Formation is defined by the Eulowrie Pyroclastics, which vary from gray, green, beige, or pink ash-rich cherty siltstone, with structures indicating fluvial or lacustrine redeposition of pyroclastic material and ignimbrite. Zircons extracted from siltstone at site 27 are a heterogeneous mixture and their varied compositions illustrate the difficulty of dating volcaniclastic rocks. They have modest contents of common Pb (Fig. 7C). Corrected onto concordia, their ages comprise a range of inherited ages and altered isotopic systems, and it is difficult to determine a primary crystallization age. The probability diagram (Fig. 7D) has four broad age components. Two ca. 370 Ma grains are clearly xenocrysts. There is a cluster of 7 grains with ages near 330 Ma, and a second cluster of 14 grains near 310 Ma. A third is defined by a continuum between 220 and 300 Ma. Only the second (310 Ma) group is geologically defensible as an emplacement age; it has an acceptable MSWD and a weighted mean age of 310.6 ± 4.0 Ma (2σ). Presumably the 330 Ma grouping is from a xenocryst component, and the tail of younger ages reflects leaked radiogenic Pb. North of Rocky Creek at The Tops, the formation contains the red Tycannah Rhyodacite, thin red ignimbrites, ash-rich siltstone, and the gray to green The Tops Rhyolite (Fig. 4). The Lark Hill Formation is disconformably overlain by Permian sediments.

Werrie Syncline

Carboniferous rocks of the lower sequence within the Werrie syncline commence with the early middle Tournaisian Tulcumba Sandstone, a cross-bedded feldspathic sandstone containing oolitic limestone lenses and abundant marine fossils; the sandstone crops out on the western limb of the syncline, where it disconformably overlies the Famennian Mandowa Mudstone. Fossiliferous shelfal mudstone, minor sandstone, and limestone, to 870 m thick, of the Tournaisian Namoi Formation conformably overlie the Tulcumba Sandstone. At the top of the Namoi Formation, above a possible disconformity, are as much as 1500 m of continental coarse feldspathic volcanolithic sandstone, con-

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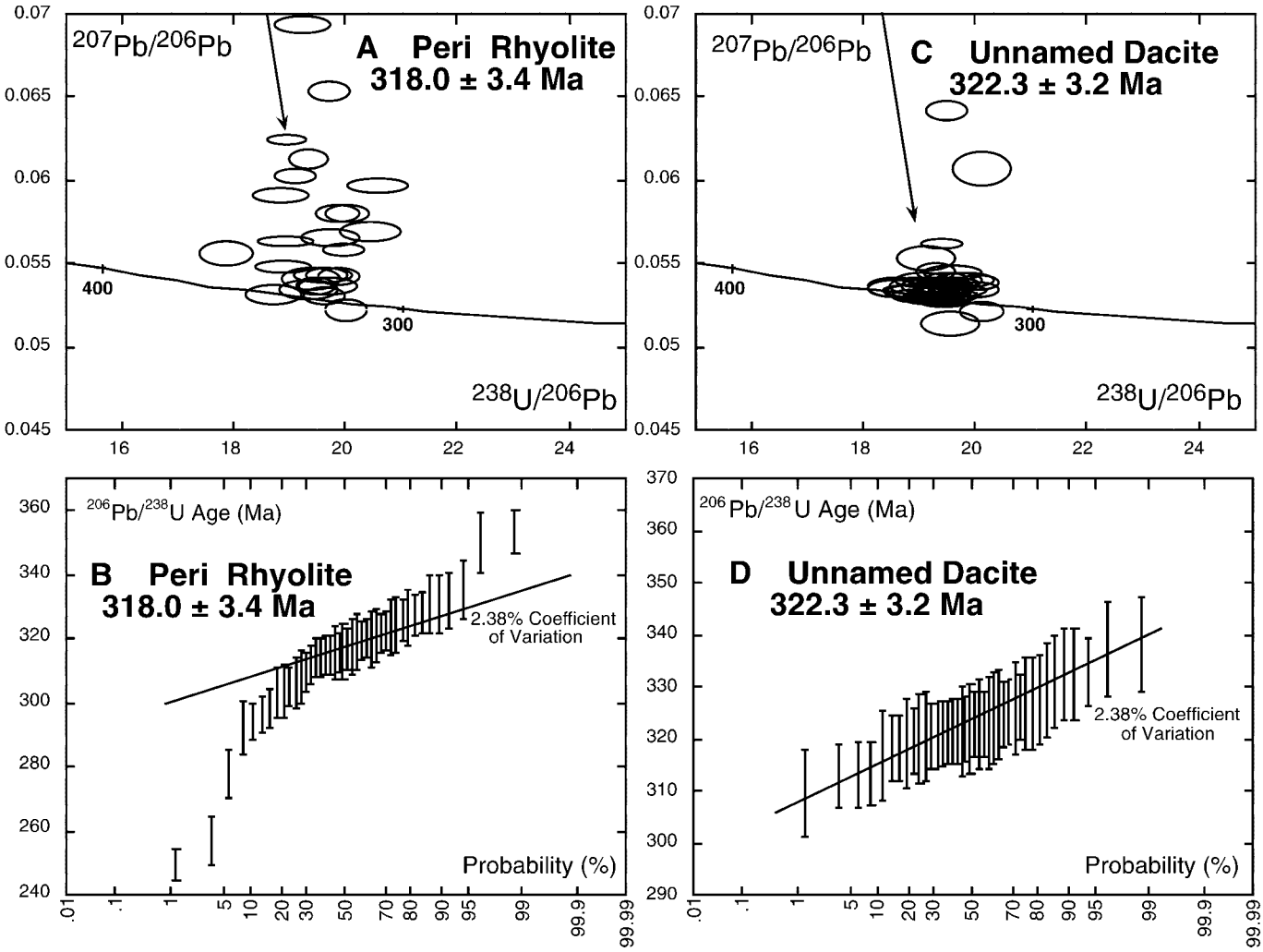


Figure 9. Concordia and probability diagrams illustrating zircon compositions in the Peri Rhyolite and an unnamed dacite in the Horton River section. Peri Rhyolite zircons have high to extreme common Pb contents and many have undergone leakage of radiogenic Pb, but an age corresponding to that of the Eastons Arm Rhyolite can be interpreted. The unnamed andesite has a very simple and homogeneous zircon population with a confidently defined age.

glomerate, heavy mineral sandstone, pyroxene andesite, and pyroclastics of the Merlewood Formation. The Kyndallyn Member, a marine incursion within the upper part of the Merlewood Formation, contains mudstone, marl, oolitic limestone, coarse gritty sandstone, and conglomerate with corals, brachiopods, and conodonts of middle Visean age (Jenkins et al., 1993; Roberts et al., 1993). Some andesites within the Merlewood Formation are sills (Moore, 1974), but according to Carey (1937), Moore (1974), and our own field observations, others are flows. One of the flows sampled (site 72) in this study gradationally overlies resedimented tuff.

Above the lower sequence, there is a disconformity representing an ~20 m.y. interval of erosion and nondeposition between the upper Visean and lowermost Westphalian. The overlying upper sequence consists of a single thick formation, the Currabubula Formation. The basal unit of the Currabubula Formation is a massive to discontinuous cobble conglomerate, followed by cross-bedded volcanolithic sandstone, conglomerate minor fluvial siltstone, rare glaciogene laminated pebbly mudstone (rhythmites), diamictite (Whetten, 1965), and abundant pyroclastics. The latter include silicic ignimbrites, the majority of which are crystal-rich, air-fall pyroclastics and cherty ash-rich resedimented siltstone. Many ignimbrites are local, but the Cana Creek Tuff, Iventure Ignimbrite, Taggarts Mountain Ignimbrite, and Piallaway Trig Ignimbrite Members are widely distributed throughout the syncline (McPhie, 1983). Irving's (1966) samples from the Currabubula Formation were taken from a position stratigraphically beneath the Canna Creek Tuff in the lower part of the formation.

The age and duration of the Currabubula Formation have been established by dating volcanic units near its base and top. The lowest (unnamed) ignimbrite in the Figtree Creek section from site 60 (Fig. 6) yielded zircons with modest proportions of common Pb and a complex array of $^{206}Pb/^{238}U$ ratios (Fig. 10). Arranged in the probability diagram (Fig. 10B), this array resolves into three age components. A single grain has a low apparent age of ca. 290 Ma interpreted as due to loss of radiogenic Pb. The main group follows the expected normal distribution, but is complicated by a tail of older ages ranging continuously to ca. 360 Ma, representing a primary crystallization age ad-

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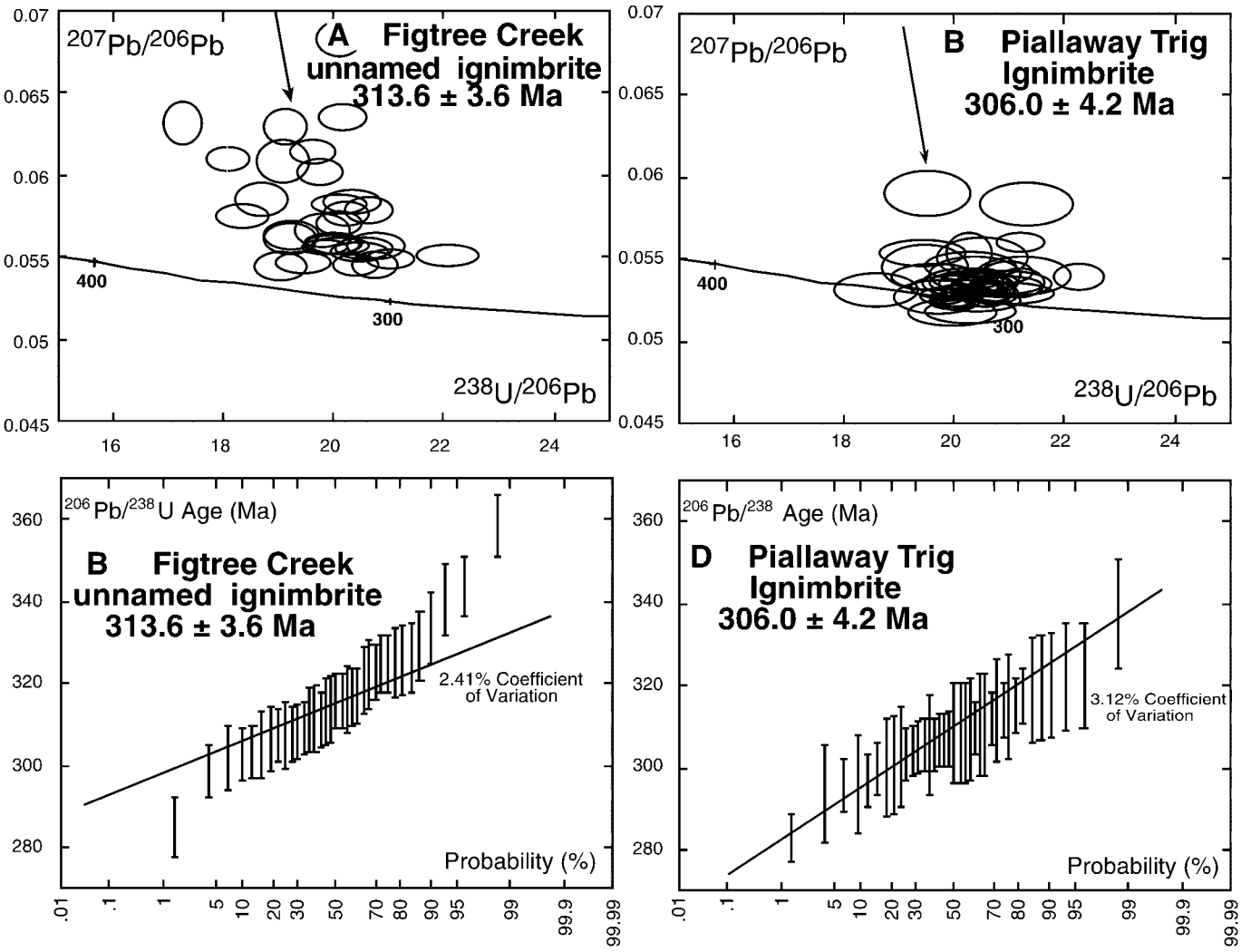


Figure 10. Concordia and probability diagrams of zircon compositions in two Currabubula Formation ignimbrites. The unnamed ignimbrite at the base of the Figtree Creek section has a magmatic zircon population complicated by admixture of older xenocrystic ages, requiring a statistical approach to recognizing the crystallization age group. Zircons in the Piallaway Trig Ignimbrite have simple and homogeneous compositions defining a single crystallization age.

mixed with older xenocrystic zircons. The xenocrystic and magmatic types of zircon cannot be distinguished morphologically. If we define those ages that depart from the main population by more than two standard deviations (Chauvenet's Criterion) as xenocrystic, then the remaining, presumably magmatic, population has an acceptable MSWD and a weighted mean age of 313.6 ± 3.6 Ma. This age could be slightly biased old or young if the purely statistical cutoff has misidentified the marginal older xenocrystic ages.

The uppermost volcanic unit, the Piallaway Trig Ignimbrite (site 73, Fig. 6), yielded a much simpler population of zircons. On the day the samples were measured, less than ideal instrumental conditions resulted in a reproducibility of $^{206}Pb/^{238}U$ ratios of >3% (usually 2%), so analytical errors are larger than normal. However, the data define a simple array of compositions (Fig. 10, C and D). All the measured ages group with an acceptable MSWD as a homogeneous population, and define a weighted mean age of 306.0 ± 4.2 Ma (2σ), which is interpreted as the crystallization age of the Piallaway Trig Ignimbrite.

Therefore, the Currabubula Formation was deposited over about 7 m.y. throughout the early to late Westphalian and possibly into the earliest Stephanian. According to McPhie (1987), it was deposited within braidplains and lakes in a piedmont region east of a volcanic chain analogous with the modern Andes Mountains of South America. Erosional events caused channel-like disconformities (McPhie, 1983) and variations in the thickness of 1000–2400 m (McKelvey and McPhie, 1985).

The Iventure, Taggarts Mountain, and Piallaway Trig Ignimbrite Members of the Currabubula Formation are mapped as far north as Gunnan Ridge (Geeve, 1995), 30 km south of the southern closure of the Rocky Creek syncline. Recent field work by Roberts (unpublished data) negates McPhie's (1983) suggestion that the Taggarts Mountain Ignimbrite extends into the Rocky Creek syncline as the Plagyan Ignimbrite. The medial position of the Taggarts Mountain Ignimbrite within the largely Westphalian Currabubula Formation suggests a younger age than is possible for the Plagyan Ignimbrite,

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a unit located at the base or within the lower part of the Rocky Creek Conglomerate.

After deposition of the Currabubula Formation there was a hiatus spanning virtually the entire Stephanian. The overlying Temi Formation and Werrie Basalt are earliest and Early Permian, respectively.

PALEOMAGNETIC PROCEDURES

In our paleomagnetic work, magnetostratigraphy was the primary objective. Hence, within the limits imposed by the occurrence of suitable lithologies, we sampled as many horizons as possible choosing the best mapped sections, hoping in this way to minimize the chances of missing short polarity chrons.

Diamictites, fine-grained sandstone and siltstone, varved shales, water-laid volcanic ash, tuff, ignimbrites, and felsic igneous rocks were sampled. At each horizon or site, three or more cores 2.54 cm in diameter were drilled and oriented using a Brunton compass. Sites were plotted on 1:40 000 air photographs and 1:25 000 topographic maps. Grid references refer to the Australian Map Grid (Universal Transverse Mercator) 1966. Geographic coordinates of sites drilled in 1995 were also obtained from the global positioning system (GPS). Stratigraphic sections were measured using a 30 m tape, compass, and clinometer, and plotted at an original scale of 1 cm to 10 m. The stratigraphic position of sampling sites is shown in Figures 4 to 6; 10–25 horizons were sampled in each section. Because formations vary rapidly along strike and contain much coarse material unsuitable for paleomagnetic work, site spacing in any one section was necessarily irregular, and in order to increase site density, multiple sections were sampled. The stratigraphic relationship of sections was known in detail from mapping, so it was possible to compile composite sections, thereby increasing the sampling density through time. Sampling density was necessarily variable, but averaged about 1 site/0.14 m.y.

Within the Rocky Creek region, the composite section was assembled from the stratigraphy of three fault blocks, from east to west, the Rocky Creek syncline, Kathrose–Maules Creek, and Darthula blocks (Figs. 2, 4, and 5). The succession in the Rocky Creek syncline block, from which the majority of samples was obtained, is based upon sections shown in Figure 4. These sections are correlated on the basis of mapped stratigraphic boundaries and recent work indicating that the Peri and Eastons Arm Rhyolites are the same unit (Xiang Wang, unpublished data). The Rocky Creek succession is represented by the right one-third of the composite section (Fig. 11).

The Kathrose–Maules Creek block is covered in its medial parts by Tertiary basalt of the Nandewar Range. The block is bounded in the north by the Kathrose and Trevallyn faults and in the south by the Kelvin and Plagyan faults. The formations within both parts of the Kathrose–Maules Creek block are the same as those in the Rocky Creek syncline block, except for the absence of the Ermelo Dacite. The Peri Rhyolite is present in the northern part of the Kathrose block but, because of lack of outcrop, cannot be stratigraphically linked with the Pine Cliffs Road section. In the Maules Creek block the Wanganui Andesite and Eastons Arm Rhyolite Members provide a link with the southern part of the Rocky Creek syncline block. Other ignimbrites within the Kathrose–Maules Creek block differ from those in the adjacent blocks. In the composite section (Fig. 11), the succession in the Kathrose–Maules Creek block is represented by the central one-third of the column and is based on data from the Pine Cliffs Road, Maules Creek, and Arizona-Barneys Springs sections (Fig. 5).

The Darthula block, sampled north of the Nandewar Range, was sampled along the Lochiel Downs–Darthula section, as shown in the left column of Figure 5. None of the ignimbrites within this block is identifiable in blocks to the east, correlation with the latter being based on the recognition of the Clifden Formation, Rocky Creek Conglomerate, and Lark Hill Formation within the Darthula block.

In the Werrie syncline, the composite section is constructed from sections measured across the eastern and western limbs of the syncline. The Merlewood and Figtree Creek sections, from the west, provide stratigraphic information on the Merlewood and lower part of the Currabubula Formation. The Washpool Creek section, from the eastern limb, extends the record to the upper part of the Currabubula Formation, but the section contains disconformities. The Canna Creek section, from a fault block on the southwestern limb, is faulted internally but provides details of part of the section missing, because of unconformities, in the Washpool Creek section. The Merlewood and Figtree Creek sections are linked by the mapped base of the Currabubula Formation. Correlations between sections within the Currabubula Formation are based on the distinctive and well mapped Canna Creek Tuff and Taggarts Mountain Ignimbrite Members (McPhie, 1983). Disconformities indicated within the composite section include some originally identified by McPhie (1983).

PALEOMAGNETIC RESULTS

In the laboratory, cores were sliced into 2.45-cm-long cylinders. Samples were measured on a three-axis cryogenic magnetometer housed in a shielded room, which reduces the intensity of the magnetic field to <400 nT. All samples were thermally demagnetized in 10 or more progressive heating steps to 570 °C or higher. The magnetizations present were then determined by line fitting to the linear portion of the demagnetization trajectories (Kirschvink, 1980). Mean angular deviation (MAD) values were generally below 5°.

Orthogonal plots of the progressive thermal demagnetization generally reveal two magnetizations designated A and B. A has low unblocking temperatures and is directed to the north and up, close to the ambient geomagnetic field (Fig. 12B). The mean is $D = 8.4°$, $I = -57.5°$, $N = 488$, $\kappa = 8$, $\alpha_{95} = 2.3°$. A third component is present at site 113 (Fig. 12F); however, this component is present only at this normal site.

The B component is unblocked at higher temperatures (unblocking temperatures typically range from 400 to 580 °C). In the upper sequence it is steeply inclined down and toward the south (Fig. 12B). This B magnetization is reversed in Paleozoic coordinates when untilted and is close to that observed by Irving (1966). It is present in rocks of all lithologies sampled in the upper sequence.

Samples from the lower sequence reveal a B magnetization representing several lithologies with moderate inclination and directed either up and to the north, or down to the south (Fig. 13, A, B, and C), normal and reverse, respectively, in late Paleozoic coordinates.

In about 15% of samples, high-temperature magnetization could not be resolved, and the paleofield could not be determined. This lack of resolution was sporadic, sometimes affecting one or two specimens from a site, sometimes all specimens from a site. This was particularly true for the Ermelo Dacite; no specimens from all eight sites drilled yielded a reliable high-temperature magnetization. For example, at site 40 (Fig. 13D) the low-temperature A component (0–300 °C) comprises 90% of the remanent magnetization and is directed close to the Earth's present field. At site 128 (Fig. 13E), however, the magnetization is directed differently at a shallow (25°) angle to the northwest (313°). At site 143 (Fig. 13F) the magnetization is directed to the north and

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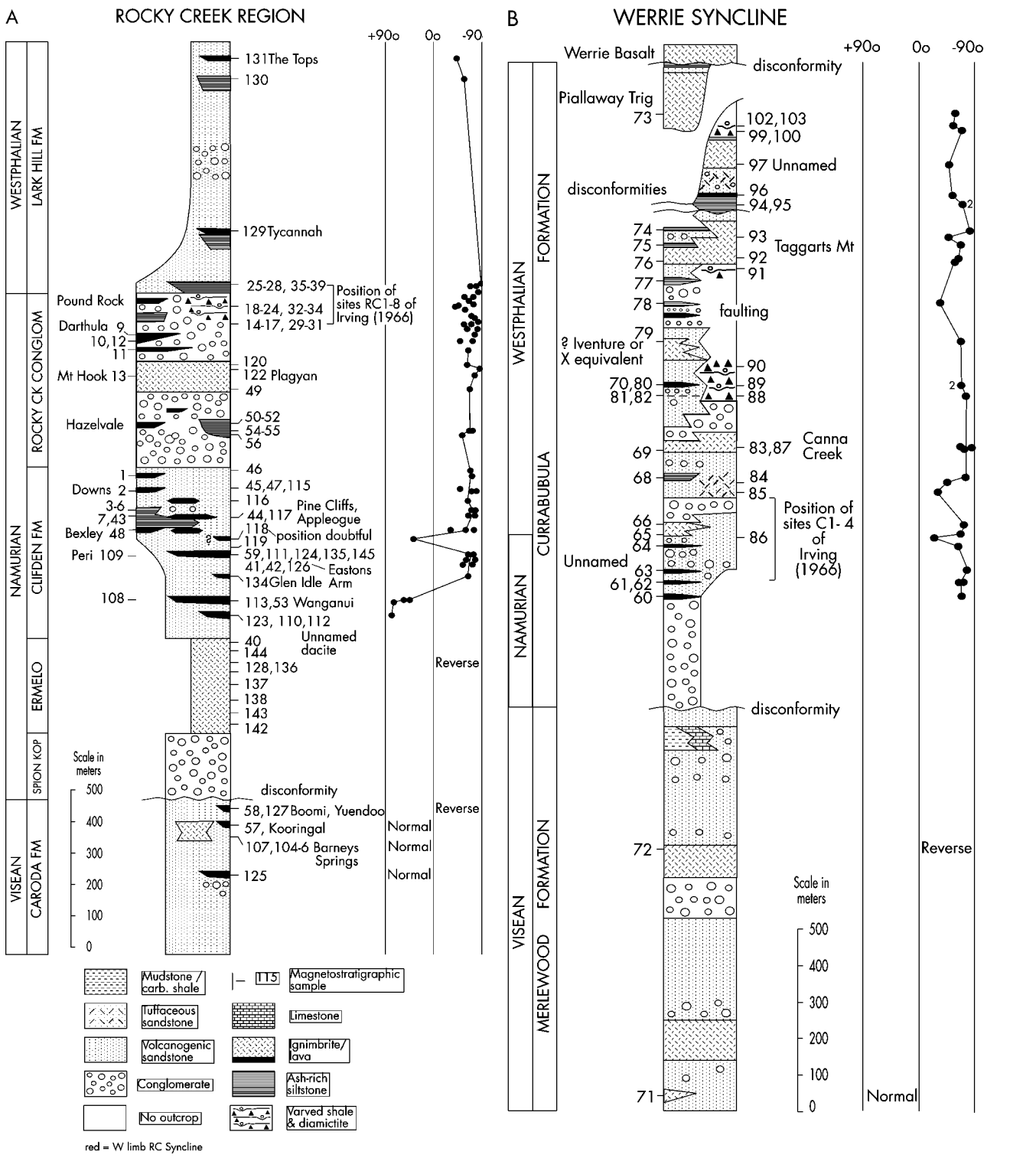


Figure 11. Composite sections of the Rocky Creek (A) and Werrie (B) synclines showing relative stratigraphic positions of the paleomagnetic sampling sites with their virtual geomagnetic poles (VGPs).

up and persists to higher temperatures, but like site 40 is close to the Earth's present field. No record of the paleofield during deposition of the Ermelo Dacite can be obtained from these data.

The B magnetizations at each site have been combined, giving specimens unit weight using Fisher (1953) statistics. Means for each site are listed after correction for tilt in Table 1. The sections are displayed in broad open folds and tilted blocks and therefore correction has been made by simple rotation about the strike.

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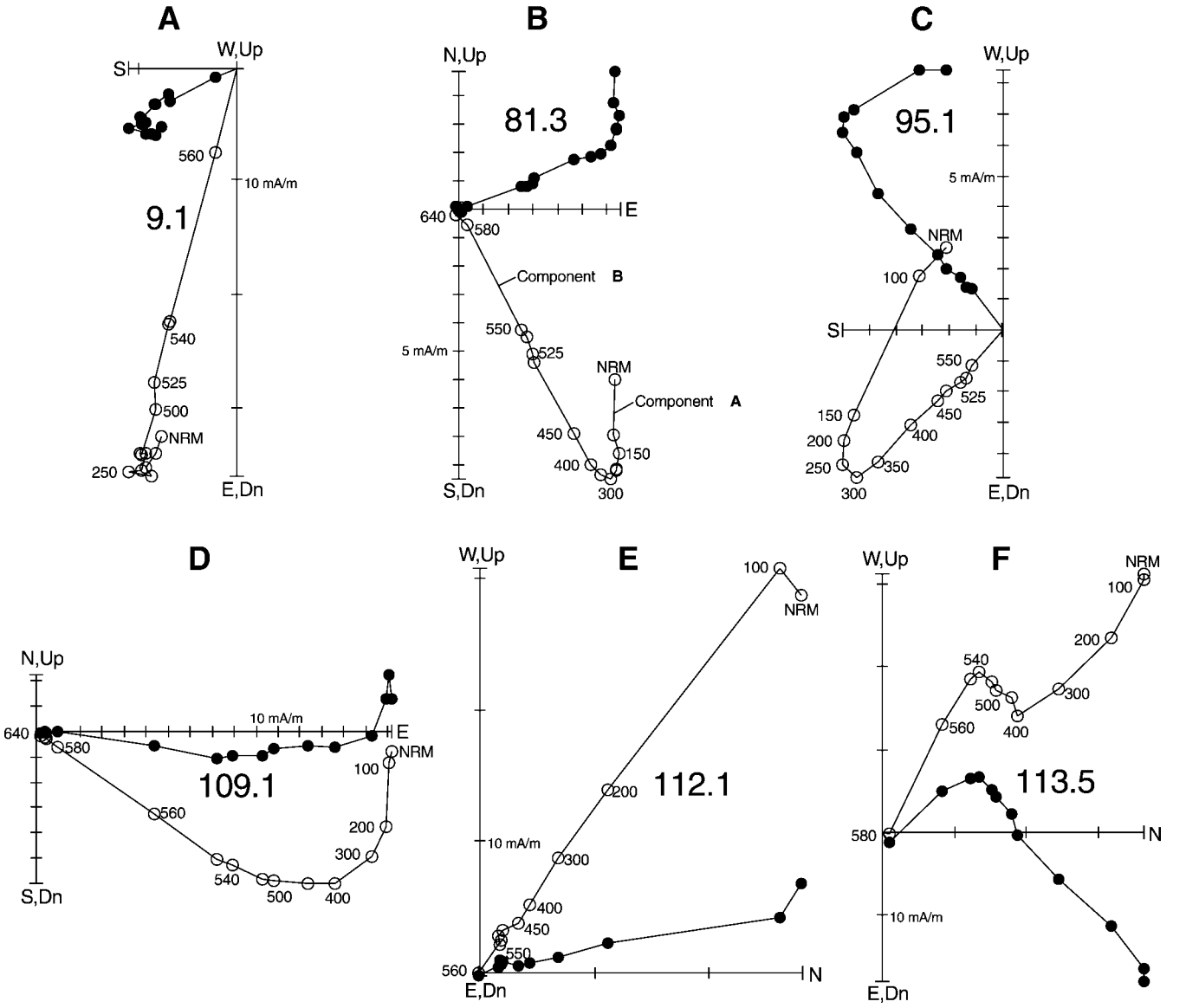


Figure 12. Orthogonal plots of thermal demagnetization from six sites from upper sequence, uncorrected for bedding tilt. A—Darthula Rhyodacite; B and C—ash-rich sedimentary rhythmites; D—Eastons Arm Rhyodacite; E—poorly welded red ignimbrite; F—Wanganui Andesite. The magnetization vectors after the heat treatment specified are plotted on two of the three possible orthogonal planes. NRM is the initial magnetization. Straight-line sections of graphs map component magnetizations A and B as shown in (Fig. 9B). Solid dots are on the horizontal plane. Open circles are on one of the two vertical planes, as indicated.

Magnetic Carrier

The unblocking temperature of B magnetization is commonly in the range of 400–580 °C, suggesting that magnetite with a wide grain-size range is the chief carrier (Fig. 12, A, D, and F). In some instances (Fig. 12D) the unblocking temperatures range above 600 °C, indicating that hematite also contributes. Within the same specimens magnetite and hematite magnetizations always have the same direction.

In an attempt to investigate further the minerals responsible for the magnetization, polished thin sections from six sites were compared with saturation isothermal remanent magnetization (IRM) acquisition curves (Fig. 14). The IRM was imposed by applying increasing magnetic fields in an impulse magnetizer.

Two specimens (sites 1 and 81) are nearly saturated by 500 and 300 mT, respectively, indicating that magnetite is the predominant magnetic mineral. Site 81, a uniformly fine grained green ash, has innumerable small opaque isotropic particles ranging in size from a few microns to smaller. The larger particles are identified as titanomagnetite. No exsolution textures, such as are common in titanium-rich titanomagnetites, were observed. We therefore inferred that magnetite is the predominant opaque mineral, and is typical for rocks of intermediate and felsic composition such as we are dealing with. Site 1, an orange-red ignim-

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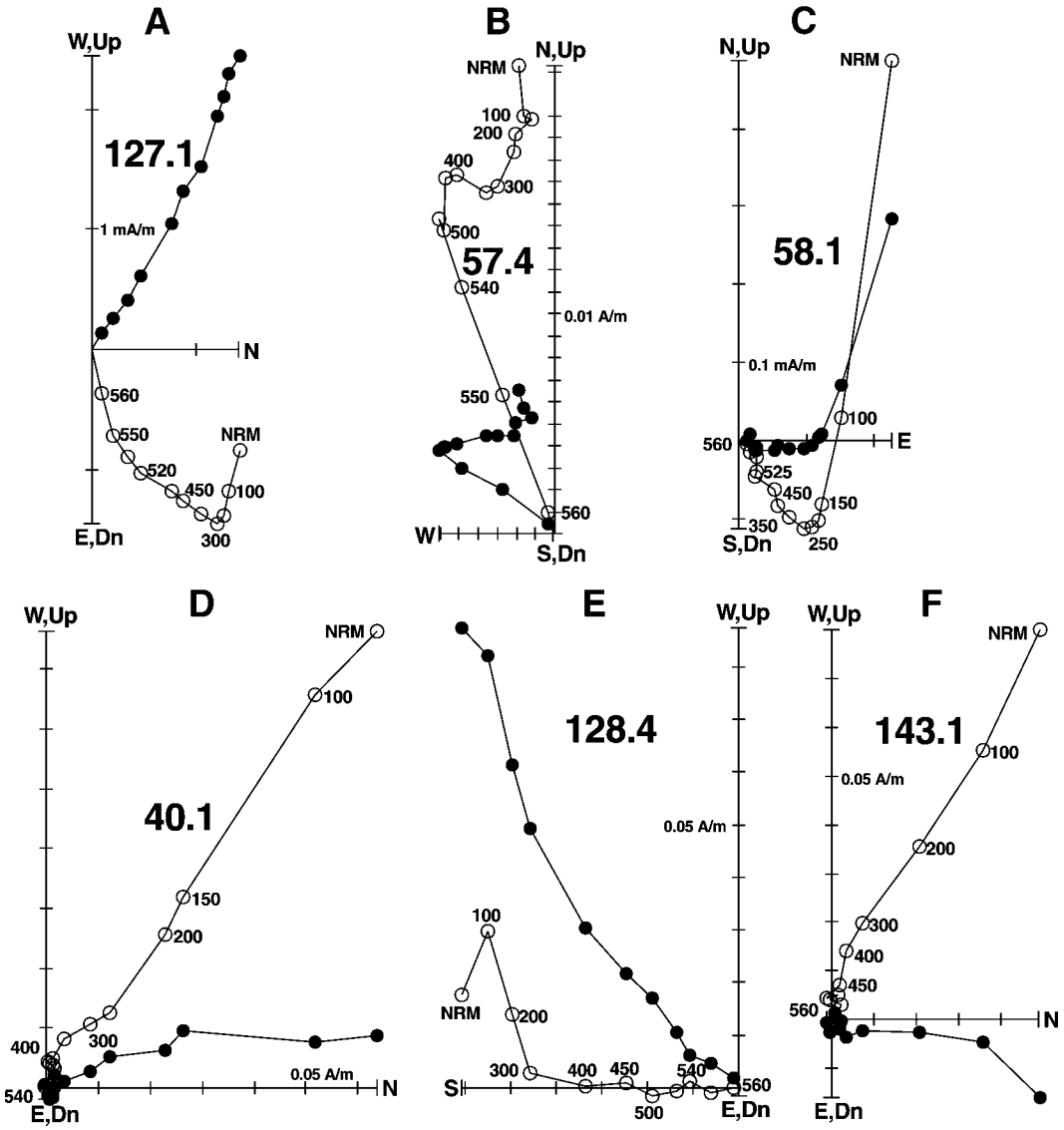


Figure 13. Orthogonal plots of thermal demagnetization from the lower sequence and Ermelo Dacite. A—Yuendoo Rhyolite; B—Kooringal Andesite; C—Boomi Rhyodacite of the Caroda Formation; D, E, F—Ermelo Dacite. All are uncorrected for bedding tilt. See Figure 7 caption.

brite, has small magnetite particles, abundant orange-tinted devitrified glass, and no identifiable hematite.

Specimens from sites 9, 13, 26, and 95 show an initial sharp rise of IRM, suggesting that magnetite predominates, but in high fields continues to rise, indicating that hematite may also be present. Sites 9 and 13 (the Darthula Rhyodacite and the Mount Hook Ignimbrite, respectively) contain blocky magnetite grains ranging from below 1 μm, and abundant large phenocrysts, originally feldspar, with red-brown centers full of disseminated hematite. Site 26, a pink-red tuff, has submicron opaque grains too small to identify, and abundant disseminated hematite. Site 95, a purple-red ash-rich strongly banded varve, has abundant fine-grained magnetite from submicron to micron size and dark red hematite pigment in the matrix.

Magnetite is the predominant magnetic carrier of B magnetization, and hematite is a subordinate carrier. Hematite, however, is common as very fine grained dust, often giving the rocks a red or orange color, and is a product of the breakdown of volcanic glass and zeolitization, which occurred soon after deposition. Much of it is probably too fine grained to contribute to the B magnetization. Without exception, hematite magnetization is parallel to that carried by magnetite, indicating that both have acquired at or soon after deposition.

Conglomerate Test

Lackie and Schmidt (1994) suggested that the Carboniferous rocks of eastern Australia have undergone extensive or perhaps complete remagnetization in the Permian, a possibility previously considered and rejected by Irving (1966). To resolve this matter, samples were taken from igneous clasts in diamictites.

In the Rocky Creek section 15 igneous clasts were drilled at site 20.

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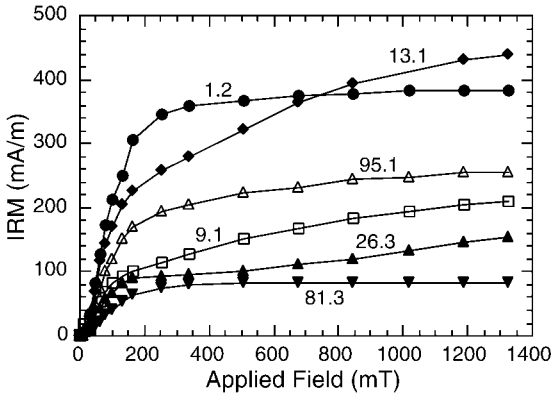


Figure 14. Initial remanent magnetization (IRM) acquisition. Site 13 is Mount Hook Ignimbrite, site 1 is unnamed red ignimbrite, site 95 is purple rhythmite, site 9 is Darthula Rhyodacite, site 26 is Eulowrie Pyroclastics, and site 81 is beige ash-rich siltstone.

Examination of thin sections confirmed that most clasts were Carboniferous, but several were probably Devonian. Orthogonal plots for three clasts are given in Figure 15. Both A and B magnetization are present. In one clast a B magnetization could not be resolved. The directions before tilt correction of the A magnetizations are close to the ambient Earth's field and are interpreted as viscous magnetizations of recent origin. The B magnetization directions, however, are random ($N = 14$, $\kappa = 2$; Fig. 16). Site 19 in rhythmites in contact with the diamictite yielded steep reversed (+85°) inclinations typical of the upper sequence.

In the Cana Creek section of the Werrie syncline 13 clasts were drilled (sites 90 and 91). Three samples from site 91 are dropstones in a rhythmite sequence. Of the clasts, 10 yield B magnetization, and only 2 have well developed A magnetizations. The B magnetizations are random ($N = 10$, $\kappa = 2$; Fig. 16). Both tests are clearly random on the Watson's (1956) criterion.

Five cores were taken from a purple mudstone directly underlying site 90. Clasts to 2 cm are embedded in the matrix. Three cores yielded B magnetizations with scattered directions, presumably because of clasts included within specimens.

These tests show that no regional remagnetization of Permian or later age has affected these rocks. It could be argued that the clasts are not representative of the rocks sampled in the upper sequence, sedimentary rocks being noticeably absent. However, it would be a coincidence beyond what we consider reasonable if regional remagnetization of sedimentary rocks had occurred and yet the igneous clasts in the same sections were unaffected, as demonstrated in Figure 16.

Slump Test

Vertically inclined strata (two cores) from a contorted sequence (Table 1) yielded southwest (210°) directions of magnetization with shallow (–5°) inclinations. A third core from a moderately inclined slumped layer yielded an inclination of 37° directed to the south. Adjacent sites in undisturbed sediments yield the usual steeply inclined reversed magnetization. The slumps are so internally contorted that restoration to the horizontal by simple rotation is not possible. These observations indicate the early acquisition of remanence in these sediments.

TILT TEST AND SUMMARY DIRECTIONS

Mean directions of the upper sequence after correction for tilt are listed in Table 2. Means have been calculated giving unit weight to each sampling site. The number of samples per site is variable (Table 1, database), and in order to focus on the better sampled sites, means have also been calculated using only those sites represented by three or more separately oriented cores. In some instances, means derived from the latter procedure have slightly higher precisions, but the differences are negligible. This shows that the data set, despite variability in sampling, is remarkably uniform.

Data from each rock formation have been subjected to an incremental tilt test. Precision parameters become maximum after untilting by amounts varying from 95% to 104%, values that do not differ significantly from 100%, as shown in Figure 17. The results are statistically the same whether all sites are used or only those with three or more samples (Table 2).

Thus the B magnetization of the upper sequence predates the major folding event in Late Permian to Triassic time. Together with the conglomerate test and the study of magnetic carriers, the tilt test results show that B magnetization records directions of the geomagnetic field at or soon after deposition during Namurian to Westphalian time.

Considering the results in more detail, entries 1–7 of Table 2 give results for each formation. Their directions are not significantly different from one another; upper sequence rocks share the same common direction. Entries 8–16 give results from the lower part of the Kiaman superchron (late Namurian to Westphalian). Igneous and sedimentary rocks are compared in entries 8–11; pyroclastic rocks, being neither the one nor the other, are not included in this comparison. The directions of igneous and sedimentary rocks are statistically identical, differing magnetization processes yielding the same results. Evidently there is no inclination error in the sedimentary rocks. Entries 17–20 give means for the Werrie and Rocky Creek synclines. They do not differ significantly, indicating that the two structures have not been rotated relative to each other.

Entries 21–24 summarize results for the upper sequence, comprising the Kiaman and the normally magnetized rocks of the lower part of the Clifden Formation. The mean inclination corresponds to a paleolatitude for this part of eastern Australia of 65.0° ± 5.6°S. Within statistical errors there is no evidence of latitudinal change during deposition of the upper sequence. Corresponding pole positions are listed in Table 3.

Lower sequence rocks of the early middle Visean, Caroda, and Merlewood Formations have B magnetization with moderate inclinations, lower than the upper sequence (Table 2). Both normal and reversed polarities are present (Table 1; Fig. 11). Regardless of sign, the directions, although scattered, are generally toward the south with moderate inclinations, with the exception of the Yuendoo Rhyolite (site 127), which is directed to the north with positive inclination. Declinations are scattered and imply relative rotations among sites; therefore we have averaged inclinations (Gaussian averages). Adding the data from three sites from the Visean Gilmore Volcanic Group (Irving, 1966) yields a mean of 52°. An inclination-only fold test yields a broad maximum after 80% untilting. It is clear that this is not a definitive result and some syn-folding magnetization may be present, but inclinations are significantly shallower in the lower than in upper sequence rocks. The inclinations correspond to a paleolatitude of about 32° during early Visean time. Evidently during the late Visean to Namurian hiatus, the paleolatitude of southeastern Australia increased, perhaps by as much as 30°.

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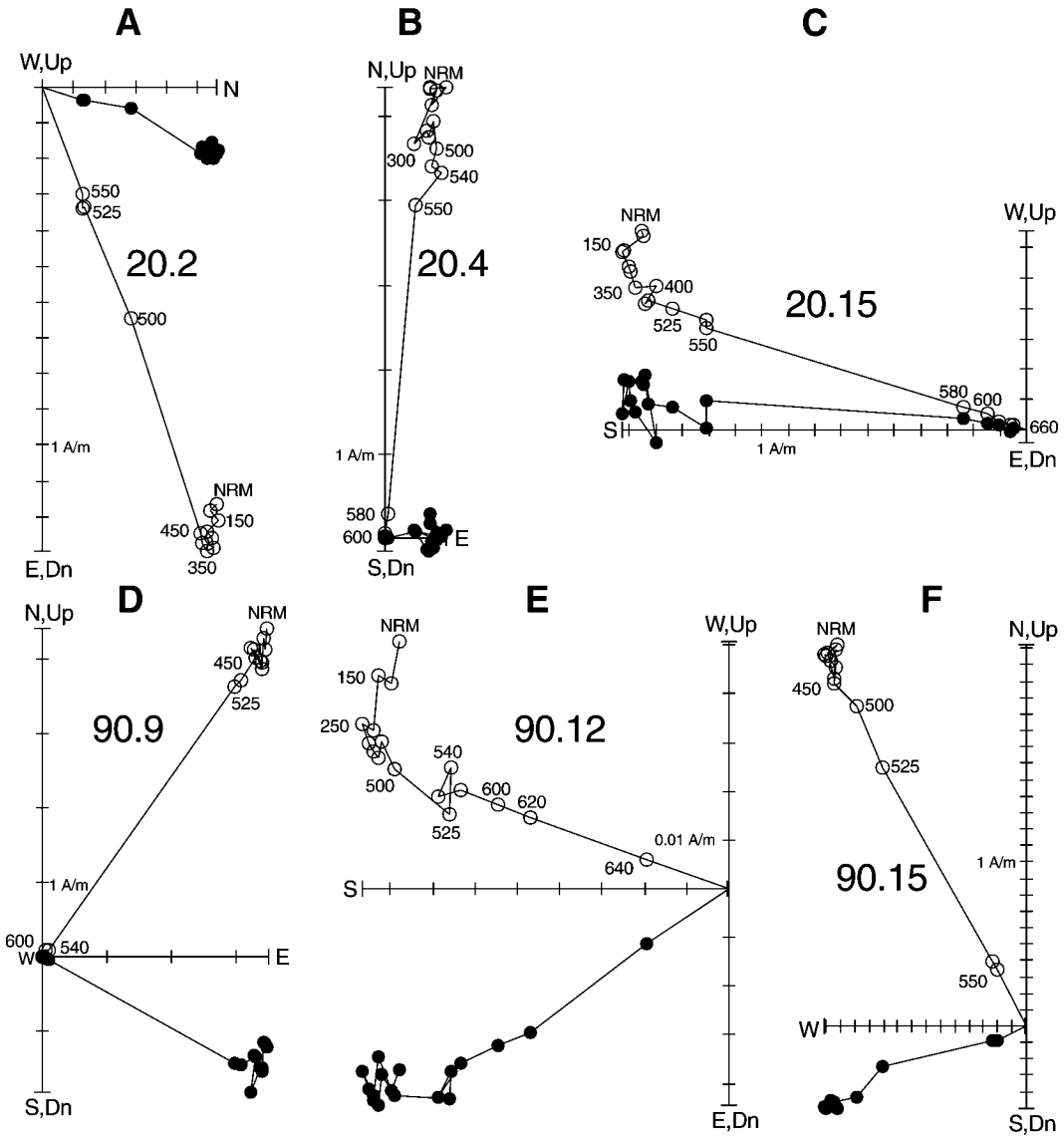


Figure 15. Orthogonal plots from diamictite clasts at site 20, Rocky Creek syncline, and site 90, Werrie syncline, uncorrected for bedding tilt. See Figure 7 caption. NRM—natural remanent magnetization.

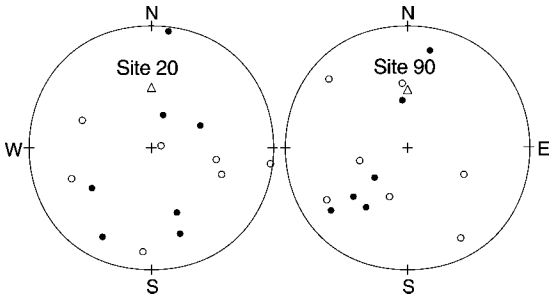


Figure 16. Directions of the B magnetization of conglomerate clasts site 20, Rocky Creek syncline, and site 90, Werrie syncline. Triangle is present Earth field.

POLARITY SUCCESSION AND GLOBAL CORRELATION OF BASE OF KIAMAN

Virtual geomagnetic pole (VGP) positions for all sites from the Rocky Creek and Werrie syncline regions were calculated with respect to the mean pole positions (Table 3). The latitudes of these VGPs are placed in stratigraphic order in the measured sections (Figs. 4–6), and compiled into composite sections (Fig. 11). The composite section in the Werrie syncline area represents about 1500 m of section and is entirely reversely magnetized. SHRIMP dates on the top and bottom are 306 ± 4.2 and 313 ± 3.6 Ma. The section, encompassing essentially the entire Currububula Formation, therefore represents about 7 m.y. in time without reversals.

The composite stratigraphic column from the Rocky Creek syncline represents more than 2000 m of section in rocks of the Clifden Formation, Rocky Creek Conglomerate, and Lark Hill Formation. All of the sites sampled in the Rocky Creek Conglomerate and Lark Hill

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TABLE 2. SUMMARY OF PALEOMAGNETIC DIRECTIONS FROM THE UPPER SEQUENCE

No.	Rock Unit	*B*	*Ds*	*Is*	*kg*	*ks*	α_{95}
1	Lark Hill Formation	7	161.1	80.9	3	45	9.1
2	Rocky Creek Conglomerate	33	185.8	78.1	9	29	4.7
3	Rocky Creek Conglomerate	23	189.2	76.6	6	33	5.4
4	Clifden Formation	34	184.5	73.7	7	22	5.4
5	Clifden Formation	29	183.4	72.7	7	19	6.2
6	Currabubula Formation	35	218.9	79.3	5	22	5.3
7	Currabubula Formation	29	224.5	77.9	6	31	4.9
8	Kiaman Superzone (igneous)	35	205.3	76.1	5	26	4.9
9	Kiaman Superzone (igneous)	33	205.6	75.8	6	24	5.2
10	Kiaman Superzone (sedimentary)	64	181.8	79.5	5	24	3.7
11	Kiaman Superzone (sedimentary)	47	184.2	79.0	6	29	3.9
12	Kiaman Superzone (include. pyroclastics)	104	191.9	78.6	5	25	2.8
13	Kiaman Superzone (include. pyroclastics)	83	194.5	77.8	6	27	3.0
14	Kiaman Superzone (Irving, 1966)	18	201.2	79.6	16	48	5.0
15	Kiaman Superzone (12 and 14)	122	193.7	78.6	5	28	2.5
16	Kiaman Superzone (13 and 14)	101	196.1	78.0	6	30	2.6
17	Kiaman Rocky Creek Syncline	69	180.7	77.4	7	30	3.2
18	Kiaman Rocky Creek Syncline	54	180.6	76.5	8	29	3.7
19	Kiaman Werrie Syncline	35	218.9	79.3	5	22	5.3
20	Kiaman Werrie Syncline	29	224.5	77.9	6	31	4.9
21	Upper Sequence (Nam. and West)	109	192.9	77.8	5	23	2.9
22	Upper Sequence (Nam. and West)	88	195.3	76.9	6	25	3.1
23	Upper Sequence (including 14)	127	194.4	78.0	5	26	2.5
24	Upper Sequence (including 14)	106	196.6	77.2	6	27	2.6

Note: B—number of sites, *Ds*, *Is*—mean directions after title correction, *kg* and *ks*—Fisher precision before and after tilt correction, α_{95}—error ($P = 0.05$). Except for 1 and 15, the entries are in pairs; the first of each pair includes all sites, the second includes only those sites with three or more samples. Entry 14 data are from Irving (1996). Entries 12 and 13 calculated using all Kiaman sites, including those in pyroclastic rocks.

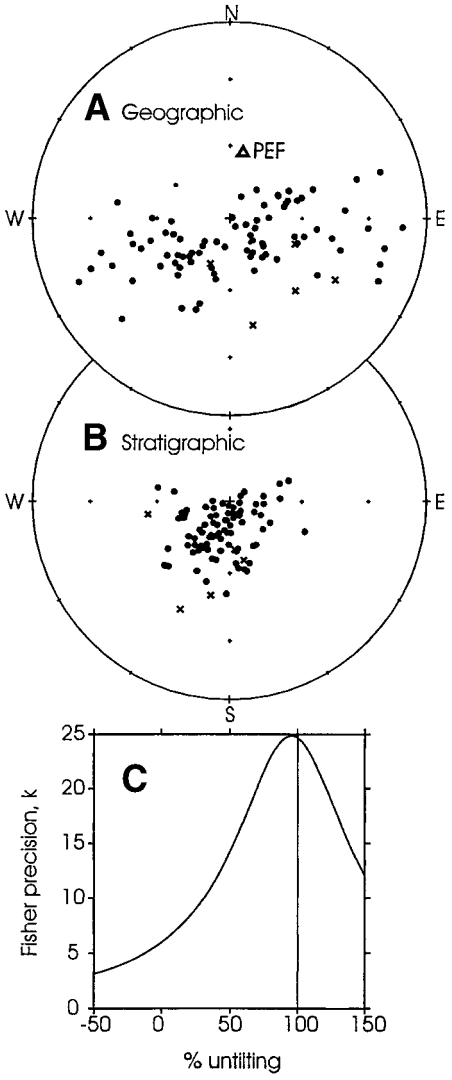


Figure 17. Site mean directions before (A) and after (B) tilt correction. (C) Precision as a function of incremental tilting for 88 sites (N > 3) from the upper sequence. Data correspond to entry 22, Table 2. PEF—present Earth field.

TABLE 3. POLE POSITIONS FOR UPPER SEQUENCE

Formation	No.	B	Lat (°S)	Long (°E)	K	$A°_{95}$
Lark Hill	1	7	46.6	157.4	14	16.6
Rocky Creek	3	23	53.9	143.7	12	9.0
Clifden	5	29	57.6	150.9	6	11.9
Currabubula	7	29	44.8	128.9	9	9.7
Mean	22	88	51.9	141.2	8	5.7

Note: No.—entry number in Table 2. *B*—number of sampling sites ($N > 2$). Latitude and longitude of south pole. K—precision of VGPs (virtual geomagnetic pole), $A°_{95}$—95% error in pole.

Formation are reversely magnetized. However, within the Clifden Formation, in the southern part of the syncline, there is a reversal between the Eastons Arm Rhyolite (317.8 ± 2.8 Ma) and the Wanganui Andesite (319.2 ± 2.8 Ma). The Wanganui Andesite is, therefore, the stratigraphically highest magnetically normal horizon to be established with confidence in Australia. The Eastons Arm Rhyolite is the stratigraphically lowest possible expression of the Kiaman in the southern part of the Rocky Creek syncline. In the northern part of that structure, however, a succession below the equivalent Peri Rhyolite contains an older reversed unit, the Glen Idle Rhyolite (site 134), which is not represented in the south. The Glen Idle Rhyolite has an extremely limited distribution and is restricted to the western limb of the syncline, where it crops out over a strike distance of only 400 m.

There is an unnamed normally magnetized red ignimbrite (site 118) located in a structurally complex region. Dip measurements and airphoto interpretation of field relationships indicate that this unit is about 50 m above the reversed Eastons Arm Rhyolite.

Other normal magnetizations that could belong to the same stratigraphic level as the inferred position of the unnamed red ignimbrite are reported from the Hunter region of New South Wales from beneath the Mount Durham Tuff Member of the Seaham Formation (Klootwijk, 1995). The Mirannie Volcanic Member from 375 m lower in the sequence has normal polarity and an age of 321.3 ± 4.4 Ma (Roberts et al., 1995a), close to that of the Ermelo Pyroclastics (321.3 ± 3.2 Ma), which is well below the onset of the Kiaman.

A clear change from normal to reverse polarity occurred ca. 318 Ma, between the normal Wanganui Andesite and the reversed Eastons Arm Rhyolite. A possible higher normal zone (site 118) is inferred and depicted in Figure 18, from the normal polarity of an ignimbrite known from only one outcrop of uncertain stratigraphic position, which would imply that the base reversal of the Kiaman could be slightly younger. Above these reversals, all observations are of reverse polarity and are within the Kiaman superchron.

In North America considerable magnetic work has been done on sediments of the Maringouin, Shepody, Claremont, and Boss Point Formations of the Cumberland basin of Nova Scotia and New Brunswick (Roy and Morris, 1983; DiVenere and Opdyke, 1990, 1991a). In Nova

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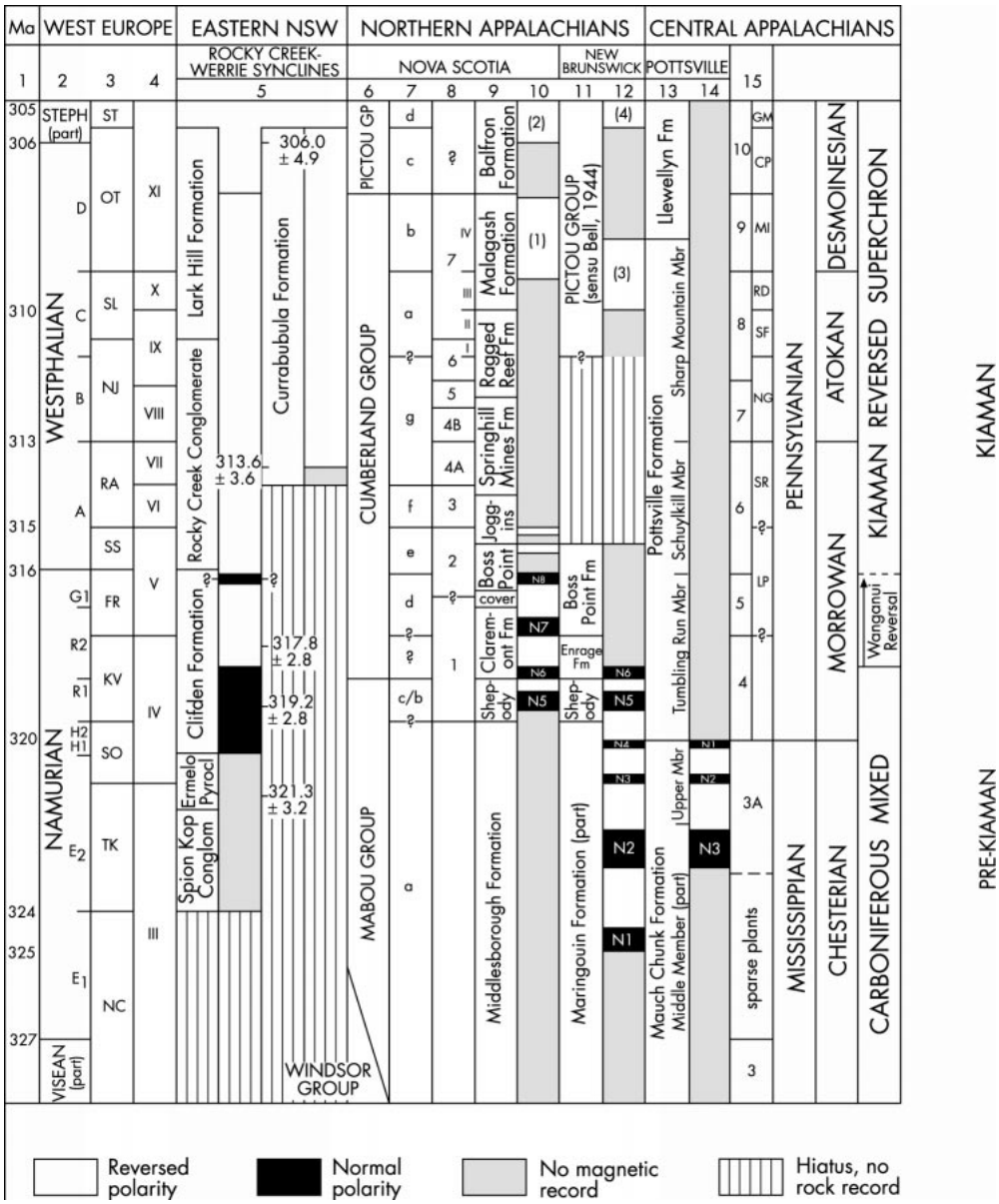


Figure 18. Correlation of the base of the Kiaman reversed superchron between Australia, North America, and Europe. Biostratigraphic and magnetostratigraphic sections of middle and Late Carboniferous rocks of northern and central Appalachians, and eastern Australia, are compared against the biostratigraphic reference scale of western Europe. The paleomagnetic record in Europe is insufficient for direct comparison of magnetostratigraphic sections. Sections indicated with no magnetic record (stippled) either have not been studied or contain no reliable record of magnetic polarity. Columns 1–15. 1—Geochronologic scale based on the Australian scale of Roberts et al. (1995a, 1995b), modified in the Namurian according to the discussion in the Appendix (see footnote 1). 2—Namurian ammonoid stages E1 (Pendleian), E2 (Arnsbergian), H1 (Chokierian), H2 (Alportian), R1 (Kinderscoutian), R2 (Marsdenian), G1 (Yeadonian) (Ramsbottom et al., 1978); Westphalian stages A (Langsettian), B (Duckmantian), C (Bolsovian), D (unnamed) (Owens et al., 1985; Riley et al., 1985). 3—Carboniferous miospore zonation of western Europe (Clayton et al., 1977). 4—Miospore assemblages in the coal seams of the Carboniferous of Great Britain (Smith and Butterworth, 1967). 5—Magnetic polarity stratigraphy for the Rocky Creek (left) and the Werrie synclines (right), eastern New South Wales (this paper). 6—Major lithostratigraphic units, Nova Scotia (Ryan and Boehner, 1994). 7—Informal palynological zonations for Maritime Provinces of Canada (Barss, *in* Belt, 1965; Barss and Hacquebard, 1967; Hacquebard, 1972, 1997). 8—Informal palynological zonations for Nova Scotia (Dolby, 1989, 1991; Yeo et al., 1988). 9—Lithostratigraphic units, Nova Scotia (Ryan and Boehner, 1994). 10—Magnetic polarity stratigraphy for the Joggins section, Nova Scotia (DiVenere and Opdyke, 1991b); post-Joggins Formation reversals. (1) Morien Group (Scotese et al., 1984); (2) Pictou Group (Roy, 1966). 11—Lithostratigraphic units, New Brunswick (Ryan and Boehner, 1994). 12—Magnetic polarity stratigraphy for the Maringouin Peninsula section, New Brunswick (DiVenere and Opdyke, 1990); post-Boss Point Formation reversals. (3) Hurley Creek Formation, Minto (Roy et al., 1968); (4) Prince Edward Island Group (Black, 1964; Roy, 1966). 13—Lithostratigraphic units, Pottsville area, Pennsylvania (Wood et al., 1969; Arkle et al., 1979). 14—Magnetic polarity stratigraphy for the Schuylkill Gap section, Pottsville (DiVenere and Opdyke, 1991a). 15 (left)—Mississippian and Pennsylvanian megafloral zones (Read and Mamay, 1964; modified by Pfefferkorn and Gillespie, 1981); (right)—Pennsylvanian palynological zones (Peppers, 1984, 1996).

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Scotia, the magnetic record is very detailed, and there are multiple normal and reversed zones, N8 being the highest normal zone, above which all strata are reversed and within the Kiaman. On these grounds, DiVenere and Opdyke (1991a) placed the base of the Kiaman at the top of normal zone N8 in the lowermost Boss Point Formation, which they regarded as Westphalian A in age. Jones (1996), however, recognizing the significance of the palynological studies of Dolby, as summarized by Ryan and Boehner (1994), suggested a late Namurian age.

The importance of placing the base of the Kiaman in a chronostratigraphic scale leads us to consider Dolby's original, and largely unpublished, work in more detail. This work is contained in open file reports (Dolby, 1986, 1987, 1988, 1989, 1991). Our correlation chart (Fig. 18) concurs with the latest report (Dolby, 1991) rather than earlier reports (Dolby, 1987, 1989), which apparently formed the basis of the summary of Ryan and Boehner (1994). However, Dolby's notion of an unconformity between the Claremont and the Boss Point Formations is not accepted, because the base of the Boss Point Formation at the type section in Nova Scotia conformably overlies and interfingers, in part, with the Claremont Formation (Ryan and Boehner, 1994). In addition, a correlation based on lacustrine flooding events indicates that the base of the Boss Point Formation is older in New Brunswick than at the type section in Nova Scotia (Plint and Browne, 1994). This correlation is in agreement with the Hacquebard (1972, his Fig. 6) recognition of two spore zones (d and e) in the Boss Point Formation in New Brunswick, but only of the younger zone (e) in Nova Scotia. Further details are provided in Appendix 1 (see footnote 1). In this paper, our revision of the stratigraphy places the N8 normal magnetozone proposed by DiVenere and Opdyke (1991b) in the basal part of the type section of the Boss Point Formation (Fig. 18), within the latest Namurian (Yeadonian, G1). The normal magnetozone is in the underlying Claremont Formation within the Marsdenian (R2).

At issue now is where and at which of the many observed magnetic reversals should the base of the Kiaman reversal be defined? The Nova Scotia magnetic record is detailed, and an uppermost unit of normal polarity (N8) can be confidently identified as that immediately below the basal Kiaman reversal. The best estimate of its stratigraphic position and age (discussed in the Appendix; see footnote 1) is latest Namurian, but it lacks direct numerical age control and there are no volcanics in the section that could be dated. The Australian data have a lower magnetic sampling density and have not detected the multiple normal zones N5, N6, N7, and N8 that precede the Kiaman in Nova Scotia. If we have placed the stratigraphic position of the possible higher Australian normal zone (above the Peri Rhyolite) correctly, then it could correlate with N6, N7, or N8. The underlying Wanganui normal zone would then correlate with N5, N6, or N7. Alternatively, if the higher normal zone in Australia is not confirmed, then the normal zone represented by the Wanganui Andesite is stratigraphically the highest and it may then be correlated with any of the normal zones N5, N6, N7, or N8 in Nova Scotia. Given the errors associated with the U-Pb ages and the uncertainty inherent in the correlation at the substage level between the Canadian Maritimes and Europe, there is sufficient uncertainty to accommodate any of these relationships.

In view of the inherent uncertainty at this detailed level of intercontinental correlation, we propose a working definition of the base of the Kiaman reversed superchron at the top of the Wanganui Andesite (319 Ma) and beneath the Peri–Eastons Arm Rhyolite (318 Ma). In Figure 18 this location corresponds to the top of N6 in Nova Scotia, and the Marsdenian Stage (R2) of the Namurian. The latest Namurian top of the N8 normal magnetic zone in the Boss Point Formation in Nova Scotia as portrayed in Figure 18 is located in a slightly younger position, and could correspond with a normal zone that may have been above the Peri Rhyolite ca. 316 Ma. These assignments are within the limits of uncertainty of SHRIMP ages and paleontological data. This uncertainty interval is indicated by a boundary zone in Figure 18. Although future work may refine its precise location more closely, we refer to the beginning of the Kiaman reversed superchron as the Wanganui reversal, after the Australian locality where the onset of uniform reversed magnetism is dated within definite limits for the first time.

CONCLUSIONS

1. A revised base for the Permian-Carboniferous reversed superchron, or Kiaman, is now recognized within successions from two continents, Australia (in New South Wales) and North America (in Nova Scotia). In New South Wales a clearly definable boundary exists in the upper part of the Clifden Formation at the top of the Wanganui Andesite Member, dated as 319.2 ± 2.8 Ma. The succeeding, reversed Peri–Eastons Arm Rhyolite has an age of 317.8 ± 2.8 Ma, indicating that the age of the Wanganui reversal is about 318 Ma. In Nova Scotia, this probably corresponds to the top of the N6 normal zone in the Claremont Formation (DiVenere and Opdyke, 1991b), within strata we assign paleontologically to the late Namurian (Marsdenian, R2). A younger normal unit could be present above the Eastons Arm Rhyolite, but has a doubtful stratigraphic position and is known from only a single sampling site. If the stratigraphic position of this normal unit is confirmed, it may correlate with normal zone N8 in Nova Scotia. The age of the base of the Kiaman is therefore likely to be between ca. 318 and 316 Ma, as indicated in Figure 18.

2. Based on the paleomagnetic evidence from Pennsylvania, the Pennsylvanian-Mississippian boundary is pre-Kiaman (DiVenere and Opdyke, 1991b) and therefore older than 318 Ma. This is in line with recent estimates of the boundary that place it at 320 Ma or older (323 Ma; Gradstein and Ogg, 1996).

3. During the late Namurian and Westphalian the paleolatitude of northeastern New South Wales, Australia, was 65° ± 5.6° S, consistent with the occurrence of glacial conditions.

APPENDIX 1. MEASUREMENT OF AGES

The Namurian and Westphalian (the Upper) sequences that form the focus of this study have a rich variety of felsic to intermediate volcanic units (flows and volcaniclastics) interbedded with the sedimentary rocks, which are suited to isotopic dating. These form the basis of the age control and correlation of the magnetic stratigraphy. The volcanic rocks were sampled for both isotopic and magnetic study as part of the same field program to ensure that isotopic age control would be integrated directly with the magnetic measurements.

Isotopic dating of volcaniclastics poses a challenge because they commonly contain xenocrystic country material of mixed ages incorporated during explosive volcanism or resedimentation. Some have been altered or weathered, and there is the hazard of isotopic leakage. Dating by measurement of U-Pb isotopes in zircon crystals has been used because zircons are resistant to resetting of isotopic systems during diagenesis, alteration, and weathering. Measurements using the SHRIMP ion microprobe have enabled large quantities of isotopic data to be recorded from the least-altered zones within individual zircon crystals, which allows inheritance and Pb loss to be recognized.

The U-Pb dating procedures are those of Claoué-Long et al. (1995). Data were collected using both the SHRIMP I and SHRIMP II ion microprobes at the Australian National University in Canberra, and the SHRIMP II at Curtin University in Perth. All samples were run with the same SL13 zircon standard, and results from the three machines are indistinguishable. The ion microprobe is used to target individual zones within zircon crystals at the 20 μm scale, and a pattern of 30–50 individual age measurements is obtained. In order to eliminate any calibration variations that might arise from separate analysis, two pairs of samples important to bracketing the age of the basal Kiaman reversal within the Rocky Creek syncline were analyzed together on the same SHRIMP mount.

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These were the Peri Rhyolite and an unnamed andesite from sites 135 and 123, and the Wanganui Andesite and Eastons Arm Rhyolite sites 113 and 111.

All ages have been calculated using the half lives recommended by Steiger and Jaeger (1977). Isotopic compositions have been corrected for common Pb by assuming its composition to be average crustal Pb of the same age as the zircons (Cumming and Richards, 1976), and by using the ^{207}Pb-based calculation described by Roberts et al. (1995a). Ages have been calculated from ^{206}Pb/^{238}U ratios, which are calibrated by reference to a value of 0.0928 for the ^{206}Pb/^{238}U ratio of the standard zircon SL13. Such ages are insensitive to correction for common Pb and to any systematic discrepancy between the half lives of ^{238}U and ^{235}U. Ages calculated from the ^{206}Pb isotope are also more precise than equivalents measured from ^{207}Pb because of the low abundance of ^{207}Pb in relatively young (e.g., Carboniferous) zircons.

SHRIMP zircon U-Pb ages referenced to the SL13 standard in this way have been shown by Claoué-Long et al. (1995) to be indistinguishable from zircon U-Pb ages measured by isotope dilution. Tucker et al. (e.g., 1998) have suggested that large discrepancies exist between zircon U-Pb ages measured by SHRIMP and isotope dilution, with isotope dilution ages being significantly older; that objection can be discounted because Tucker's isotope dilution ages were calculated using ^{207}Pb instead of ^{206}Pb, and several authors have pointed out that this approach yields discrepant old ages (e.g., Kamo et al., 1996). The good agreement between ages obtained from different isotopic systems has been demonstrated from dating of the Permian-Triassic boundary: SHRIMP zircon U-Pb dating normalized to the SL13 standard (Claoué-Long et al., 1991), ^{40}Ar/^{39}Ar dating of sanidine feldspar in the same horizon (Renne et al., 1995), and isotope dilution zircon U-Pb dating (Bowring et al., 1998) have all yielded statistically identical results. The SHRIMP-based ages reported here are, therefore, equivalent with other well-measured zircon ^{206}Pb/^{238}U ages, and are comparable with ^{40}Ar/^{39}Ar ages, a point that becomes important when linking these data to the international Phanerozoic time scale.

The principal uncertainty in age measurement, whether by SHRIMP, isotope dilution, or ^{40}Ar/^{39}Ar, is in determining the presence of inherited age components and age components arising from alteration. In order to do this grain by grain (and this is one of the advantages of the SHRIMP technique), large quantities of within-grain spot data were collected from individual grains. Two diagrams are presented for each sample. Raw isotopic compositions, without correction for common Pb, are displayed in the Tera and Wasserburg (1974) type of Concordia diagram, which displays directly the ^{206}Pb/^{238}U ratio from which ages are derived, and the raw ^{207}Pb/^{206}Pb measurements that emphasize the varying proportions of common Pb. In these diagrams, inheritance and Pb loss result in scatter subparallel to the ^{206}Pb/^{238}U axis, which is the most reliably measured value in SHRIMP zircon analyses. The cohesion of ^{206}Pb/^{238}U ages is assessed further by ranging the data on a probability plot, which has the effect of normalizing the data to the ideal normal distribution, which appears as a straight line of slope proportional to the standard deviation. The reference standard deviation plotted on each probability diagram is the observed reproducibility of measurements of the standard zircon SL13, which were alternated with measurements of unknown zircons in the sample.

APPENDIX 2. GLOBAL CORRELATIONS OF BASAL KIAMAN TIME, BIOSTRATIGRAPHY, AND MAGNETOSTRATIGRAPHY

The purpose of this Appendix is to explain the interpretations that have guided the global correlations of stratigraphy illustrated in Figure 18.

Column 1. Geochronologic Scale

The international Carboniferous time scale is reasonably well established in the uppermost Carboniferous (Westphalian and above) and in the Early Carboniferous below the Visean. Calibration of the Namurian Chron, during which the onset of the Kiaman Reversed Superchron took place, is less certain because of conflicting isotopic ages, frustrating confident correlations between continents.

The time column is based on the Carboniferous time scale of Roberts et al. (1995a, 1995b), which is calibrated in the latest Carboniferous by the European ^{40}Ar/^{39}Ar sanidine ages of Lippolt and Hess (1985) and Hess and Lippolt (1986), and in the Early Carboniferous by zircon U-Pb dating of Australian brachiopod zones. Correspondence of the Australian and European dating is established at the Westphalian B/C boundary, where the European Z tonstein is dated at 311 Ma by both isotopic techniques (Claoué-Long et al., 1995).

In the Namurian, which contains both the Middle Carboniferous boundary and the onset of the Kiaman Superchron, there are no isotopic ages available close to the stratotype of the Middle Carboniferous boundary at Arrow Canyon, Nevada. The Middle Carboniferous boundary is defined by the first appearance of the conodont *Declinognathodus noduliferus*, which is recognized in northern England at about the *Eumorphoceras-Homoceras* ammonoid transition, near the base of the Chokierian (H1) stage (Varker et al., 1991). The Carboniferous time scale of Jones (1996) estimated the boundary age as ca. 314 Ma, guided in part by the reference in Riley et al (1995) to zircon dating of the E2a3 and E2b2 Arnsbergian marine bands at 314.4 ± 4.6 and 314.5 ± 4.6 Ma. These two ages are not now considered reliable and should not be used to fix the time scale. Their use distorted the Carboniferous scale of Jones (1996) by dividing the Middle Carboniferous stages into grossly unequal durations (about 13 m.y. for the Pendleian and Arnsbergian stages, and about 3 m.y. for all of the Chokierian, Alportian, Kinderscoutian, Marsdenian, Yeadonian, Langsettian, and Duckmantian stages). Instead, we suggest an age for the Middle Carboniferous boundary near 320 Ma, consistent with the European sanidine ^{40}Ar/^{39}Ar ages of Namurian tuffs in the Ostrava Formation, Czech Republic (Lippolt and Hess, 1985; Hess and Lippolt, 1986), for the Poruba (319.5 ± 7.8 Ma), and the Jaklovec (324.6 ± 8.0 Ma) Members. The Arnsbergian goniatite subzones E2a and E2b are present in the uppermost marine bands of the Jaklovec and Poruba Members (Dvorak, 1994). In this Namurian calibration, we concur with the reasoning in Scale B of Menning (1997).

The Visean-Namurian boundary must be older than the sanidine age of the Jaklovec Member (ca. 325 Ma), and from Australian data, is taken at ca. 327 Ma (Roberts et al., 1995a, 1995b). It is fixed by zircon U-Pb ages from the Southern New England orogen of New South Wales, and lies between the late Visean Paterson Volcanics (328.5 ± 1. 4 Ma) and the Namurian Johnsons Creek Conglomerate (323.2 ± 3.2 Ma; 321.9 ± 4.0 Ma). The Johnsons Creek Conglomerate overlies the McInnes Formation, which contains the *Spelaeotriletes ybertii* Assemblage. As the first appearance of monosaccate pollen in the earliest Namurian (Clayton et al., 1991) defines the base of the *Spelaeotriletes ybertii* Assemblage, the Johnsons Creek Conglomerate lies above the Visean (Roberts et al., 1995a).

Column 5. Eastern New South Wales

The lack of fossils in the magnetic polarity succession of the Rocky Creek and Werrie synclines does not permit us to place the Kiaman Reversal directly into a global biostratigraphic context. In the absence of biostratigraphic data, the Carboniferous sequence must be calibrated to the SHRIMP zircon (^{206}Pb/^{238}U) ages and their error margins, as determined in this paper. Consequently, Figure 18 shows a "best fit" correlation for the Kiaman Reversal, and an indirect conceptual correlation of the eastern New South Wales sequence to the West European and North American palynological and megafloral zones.

Columns 6–12. Northern Appalachians

Nova Scotia (Columns 6–10)

The lithostratigraphic nomenclature used here follows that of the Nova Scotia Department of Mines and Energy (Ryan and Boehner, 1994). Further details of the Carboniferous of Nova Scotia are given in the latest synopsis by Gibling (1995).

Several palynostratigraphic schemes have been used to correlate the Carboniferous coalfields of the Canadian maritime region, the details of which remain mostly unpublished (Hacquebard, 1997). The provisional zonation originally proposed for the Canso, Riversdale, and Cumberland Groups (Barss, *in* Belt, 1965) was later extended to include the overlying Pictou Group (Barss and Hacquebard, 1967). The zones were slightly revised by Hacquebard (1972, Tables 4 and 5), who depicted their distribution in sections throughout the Fundy basin from New Brunswick to Newfoundland (1972, Fig. 6). This zonation consisted of six spore zones (in ascending order, a, c/b, d, e, f, and g) for the Mabou (Canso) and Cumberland (including Riversdale) Groups, and four zones (a, b, c, and d) for the late Westphalian and Stephanian parts of the overlying Pictou Group (sensu Bell, 1944). Overviews of this earlier work are given by Utting (1987) and Hacquebard (1997). In the late 1980s, Dolby studied the palynology of the Late Carboniferous sequence of Nova Scotia, the results of which are contained in a series of open-file reports (1986–1989, 1991). Brief summaries of the results of this work have been published by Ryan et al.(1990), Ryan and Boehner (1994), and Yeo et al. (1988).

In Figure 18, the correlation of the Nova Scotia palynological zonation of Dolby with Britain (Smith and Butterworth, 1967) and western Europe (Clayton et al., 1977) is based on the latest of Dolby's unpublished open-file reports (Dolby, 1991), but slightly modified to show an alternative correlation for the base of the Boss Point Formation (see following). A tentative attempt is made

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to correlate the Dolby (1991) zones with those of Hacquebard (1972, 1997) for the Carboniferous coalfields of the Canadian maritime region and of Peppers (1984, 1996) for the Pennsylvanian of the United States. The palynological zones of the United States are correlated with the Late Mississippian and Pennsylvanian megafloral zones (Read and Mamay, 1964; modified by Pfefferkorn and Gillespie, 1981).

Dolby (unpublished: 1987, 1989, 1991) recognised seven informal assemblage zones that can be traced throughout the Cumberland basin. Six of these zones are present along the Joggins shoreline, from Downing Cove to Spicers Cove, where they range in age from late Namurian to early Westphalian C. They were numbered 1 to 7 by Dolby (unpublished: 1991), and named but not defined in the summary account given by Ryan and Boehner (1994), which appears to be based on Dolby's (1987, 1989) earlier unpublished work. Some of the zones were later redefined by Dolby (unpublished: 1991), and therefore, the zones are unnamed in Figure 18 and in the brief accounts given below, pending the publication of Dolby's studies. Ryan and Boehner (1994, enclosure 1) depicted the positions of the zones within the Mabou and Cumberland Groups, and their correlation with the standard Late Carboniferous (Silesian) succession of western Europe.

Zone 1 is represented at Downing Cove in the lower part of the Joggins section, and contains the diagnostic species *Potonieisporites* spp., *Crassispora kosankei, Ibrahimisporites magnificus, Colatisporites denticulatus,* and *Rugospora* cf. *corporata* in the Shepody and Claremont Formations (Dolby, 1991). It also contains many species in common with the Windsor-Canso Boundary Beds (SM Zone) of Utting (1987), which were originally thought to be reworked in the basal Boss Point Formation and Claremont Formation (Ryan and Boehner, 1994), but later accepted as in situ, as their earlier extinction in the Visean of Europe was regarded as climate related. Dolby (1991 [unpublished]) correlated the zone to the late Namurian KV Zone of Clayton et al. (1977), and the *Crassispora kosankei* Assemblage IV of Smith and Butterworth (1967). However, the base of the zone (and the base of the c/b Zone) may possibly extend into the SO Zone of Clayton et al. (1977). Pivotal to the argument is the upper limit of the range of *Grandispora spinosa* (Asbian–Chokierian), a species typical of the c/b Zone, also recognized in the Shepody Formation (Hacquebard, 1972), as it last appears just above the Middle Carboniferous boundary (H1b1) in northern England (Varker et al., 1991; Turner et al., 1994). There is no evidence fixing the base of the c/b Zone from the upper limit of the a Zone, as the latter is present in the Windsor Group and ranges in age from middle Visean to an undefined position within the early Namurian.

The upper limit of Zone 1 is also contentious. A sample from the base of the Boss Point Formation was originally included by Dolby (1987) in his Zone 1, based on the lack of a *Florinites* palynoflora and the Windsor-Canso (SM) assemblage of Utting (1987). He later suggested that this sample was possibly from Zone 2, but with a restricted spore composition, and consequently Zone 1 and Zone 2 coincided with the Claremont and the Boss Point Formations, respectively. We do not question this revision, but argue against his notion of an unconformity between these formations (see foregoing text). Dolby (1991) implied a latest Namurian hiatus (Namurian C; Yeadonian Stage), suggesting that the equivalent of the FR Zone of Clayton et al. (1977) is missing. We propose an alternative interpretation, one that accepts Zone d as equivalent of the FR Zone, the upper part of Zone 1, the base of Zone 2, and the lower half of the *Densosporites anulatus* Assemblage V of Smith and Butterworth (1967). *Potonieisporites elegans,* a diagnostic species of Zone d, is at the upper limit of its range in the Namurian C (Yeadonian), and supports the notion that Zone d is a correlate of the FR Zone. It is also present in the French Lick Coal of Indiana, which Peppers (1996) places in Namurian C.

Zone 2 is present throughout the Boss Point Formation, and extends into the lower part of the Joggins Formation. Dolby recorded the diagnostic species *Spelaeotriletes triangulus/arenaceous, Florinites* spp., *Reticulatisporites polygonalis,* and *Savitrisporites* 53/12. As *Spelaeotriletes triangulus/arenaceous* dies out within the SS Assemblage Zone of Clayton et al. (1977), Zone 2 is early Westphalian A in age, and correlates with the upper part of the *Densosporites anulatus* Assemblage V of Smith and Butterworth (1967). These assemblages, in turn, can be correlated with Zone e (Hacquebard, 1972) and the upper LP Assemblage Zone of Peppers (1984, 1996).

Zone 3 is present in the upper part of the Joggins Formation and the basal part of the Springhill Mines Formation. Dolby (1991) recorded the diagnostic species *Punctatosporites* spp., *Lophotriletes* '*spinosaetosus*', *Radiizonates* spp., *Savitrisporites* 53/12, and? *Vestispora* 8901, which he correlated with the lower part of the RA Assemblage Zone of Clayton et al. (1977) and the *Schulzospora rara* Assemblage VI (Smith and Butterworth, 1967), of late Westphalian A age. The first appearance of *Florinites medipudens* in Zone 3 also supports a correlation with Assemblage VI, and with the lower SR Assemblage Zone of Peppers (1984, 1996). Zone 3 can probably be correlated with Zone f in Hacquebard (1972).

Zone 4A is present in the lower (not basal) part of the Springhill Mines Formation. Dolby (1991) listed the diagnostic species *Vestispora tortuosa, Radiizonates* spp., *Savitrisporites* 53/12, *Punctatosporites* spp., *Cananoropollis mehtae,* and *Lophotriletes* '*spinosaetosus*'. He correlated the zone with the *Schulzospora rara* Assemblage VII (Smith and Butterworth, 1967) with the uppermost part of the RA Assemblage Zone of Clayton et al. (1977), of latest Westphalian A age, which, in turn, is correlated with the SR Assemblage Zone of Peppers (1984, 1996). The base of Zone 4A is correlated with the base of Zone g in Hacquebard (1972) at the first appearance of *Vestispora tortuosa* in Zone 4A and, under its junior synonym *V. cancellata* (Smith and Butterworth, 1967), in Zone g.

Zone 4B is present in the upper (not uppermost) part of the Springhill Mines Formation. Dolby (1991) listed the diagnostic species *Punctatosporites* spp., *Lophotriletes ibrahimii,* and *Apiculatisporis variocomeus,* which he assigned to the lower part of the *Dictyotriletes bireticulatus* Assemblage VIII (Smith and Butterworth, 1967) of earliest Westphalian B age. The base of Zone 4B is also correlated with the base of both the NJ and NG Assemblage Zones (Clayton et al., 1977; Peppers, 1996).

Zone 5 is present in the uppermost Springhill Mines Formation and the lower part of the Ragged Reef Formation. Dolby (1991) listed the diagnostic species *Florinites junior* and *Punctatosporites* spp., which he correlated with the upper part of the *Dictyotriletes bireticulatus* Assemblage VIII (Smith and Butterworth, 1967) of middle Westphalian B age. Zone 5 may also be correlated to the NJ and NG Assemblage Zones (Clayton et al., 1977; Peppers, 1984, 1996).

Zone 6 is present at the top of the Joggins section in the Spicers Cove Member of the Ragged Reef Formation (220 m), where it probably straddles the Westphalian B-C boundary. Dolby listed the diagnostic species *Vestispora pseudoreticulata, V.* cf. *magna, Florinites junior, Microreticulatisporites nobilis, Sinuspores sinuatus,* and *Punctatosporites* spp. The lower boundary of Zone 6 can be correlated with the base of the *Vestispora magna* Assemblage IX (Smith and Butterworth, 1967), i.e., within Westphalian B and within the NJ and NG Assemblage Zones (Clayton et al., 1977; Peppers, 1984, 1996).

Dolby's Zone 7 is recognized by the first appearance of a new bisaccate pollen species, probably belonging to *Illinites.* It is not included within the Joggins section, but is present in other parts of the Cumberland basin, in the Malagash and Balfron Formations (i.e., Pictou Group sensu Bell, 1944), which correlate with Westphalian C-D. The lower boundary of Zone 7 is probably within the *Vestispora magna* Assemblage IX (Smith and Butterworth, 1967), i.e., within Westphalian C, and may correlate with the base of the SL Assemblage Zone (Clayton et al., 1977). The upper part of the zone is correlated with the *Thymospora obscura* Assemblage XI (Smith and Butterworth, 1967), i.e., Westphalian D, and with the OT Assemblage Zone (Clayton et al., 1977).

Dolby (*in* Yeo et al., 1988) recognised four provisional zones in the Pictou Coalfield of the Stellarton basin. He referred his Zones I, II, and III to Westphalian C, and Zone IV, defined by the first appearance of *Cadiospora magna* and *Thymospora pseudothiessenii,* to Westphalian D. In terms of the earlier zones of the Pictou Group (sensu Bell, 1944), established by Barss and Hacquebard (Hacquebard, 1972), Dolby (*in* Yeo et al., 1988) tentatively correlated his Zones I to III with Zone a, and his Zone IV with Zone b. In this paper we have taken Dolby's Zone I as earliest Westphalian C and the upper part of his Zone 6; his remaining Zones II to IV we take as equivalent to his Zone 7 (Fig. 18).

Two further zones of reversed magnetic polarity above the Boss Point and Joggins Formations are shown in Figure 18 (Column 10), viz., within the Pictou Group (Roy, 1966) and the Morien Group (Scotese et al., 1984), for which we concur with the ages given by Gibling (1995), van de Poll (1995), and Zodrow and Cleal (1998).

New Brunswick (Columns 11 and 12)

The lithostratigraphic group names from Nova Scotia (Ryan and Boehner, 1994) are used instead of those from New Brunswick (Hopewell Group, Riversdale Group) for the Maringouin Peninsula. This means the Mabou Group includes the Maringouin and Shepody Formations, and the Cumberland Group includes the Enragé (= Claremont) and Boss Point Formations (previously included in the Hopewell Group and Riversdale Group, respectively). For an overview of the Carboniferous of New Brunswick, the reader is referred to Gussow (1953), and the latest synopsis by van de Poll (1995).

The magnetostratigraphic section of the Maringouin and Shepody Formations was sampled by DiVenere and Opdyke (1990) from the south-dipping limb of the Maringouin anticline on the eastern shore of the Maringouin Peninsula. The

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Maringouin Formation is an unfossiliferous red-bed sequence, which Gussow (1953) referred to the Canso Group because of its conformable and gradational relationship with the overlying Shepody Formation. Thus, the Maringouin Formation is correlated with the Middleborough Formation of Nova Scotia and has a minimum age of early Namurian. The basal contact of the Maringouin Formation with the underlying Windsor Group was not seen. The Shepody Formation contains megafloral remains, nonmarine bivalves, and conchostracans (Copeland, 1957), all of which indicate a Namurian age. Although the Enragé Formation overlies the Shepody Formation with angular unconformity in the Dorchester area, the contact between the formations is conformable at Cape Enragé and Cape Maringouin. At the latter locality, the Enragé Formation is about 70 m thick and is the distal margin of the thick sequence (over 1500 m thick) of fanglomerate and arkosic sandstone of the Claremont Formation of Nova Scotia. Hacquebard (1972, Fig. 6) does not depict any palynostratigraphic data available for the Maringouin, Shepody, and Enragé Formations; the first is in the Boss Point Formation, where Zone d is overlain by Zone e at the Westphalian A boundary. As mentioned above, this zonal relationship, when compared with the Joggins section, indicates that the base of the Boss Point Formation is older in New Brunswick than in Nova Scotia.

Two further intervals of reversed magnetic polarity above the Enragé Formation are indicated in Figure 18 (Column 12), viz., the Hurley Creek Formation of the Minto area (Roy et al., 1968), and the Prince Edward Island Group (Black, 1964; Roy, 1966). Hacquebard (1972, Fig. 7) depicted the Hurley Creek Formation containing Zone b of the Pictou zonation, which is usually correlated within Westphalian C (van de Poll, 1995). However, Dolby (*in* Yeo et al., 1988) tentatively correlated Zone b with his Zone IV and Westphalian D (Fig. 18). Thus, we place the Hurley Creek Formation in Figure 18 straddling the Westphalian C/D boundary. We concur with the position of the Prince Edward Island Group within the Stephanian.

Columns 13–15. Central Appalachians

Pottsville, Pennsylvania (Columns 13–15)

DiVenere and Opdyke (1991a) sampled the upper part of the Mauch Chunk Formation in the original ''type area'' of the Pennsylvanian in the Southern Anthracite field at Schuylkill Gap, south of Pottsville. At this locality, the contact between the upper member of the Mauch Chunk and the overlying type section of the Pottsville Formation defines the Mississippian-Pennsylvanian boundary (Wood et al., 1969). In the northern part of the area, the upper part of the Mauch Chunk Formation interfingers with the Tumbling Run Member of the Pottsville Formation, which contains the youngest of Read and Mamay's (1964) Pennsylvanian floral zones (Zone 4, *Neuropteris pocahontas* and *Mariopteris eremopteroides*), and, therefore, straddles the Mississippian-Pennsylvanian boundary (Wood et al., 1969).

At Schuylkill Gap, where the top of the Mauch Chunk Formation is defined arbitrarily at the top of the uppermost red beds, only 340 m of the total thickness (1159 m) of the formation is exposed (Wood et al., 1969). The virtual absence of animal fossils and identifiable fossil plant remains in the Mauch Chunk Formation provides no direct paleontological evidence of its age. The base of the formation is probably no older than late Visean (III), as Read and Mamay (1964) assigned the plants in the upper part of the Pocono Formation to their *Triphyllopteris* Zone 2, which ranges from late Tournaisian to late Visean (III) in age (Wagner, 1984).

Jennings (1985) descirbed the plant *Adiantites antiquus* from 300 m below the top of the Mauch Chunk Formation, near the base of the upper exposed part of the middle member, and the base of the magnetostratigraphic section of DiVenere and Opdyke (1991a). He implied a Visean age for the presence of *A. antiquus* at this site, based on its occurrence in Europe, but noted other reports from ''a few scattered sites identified as lower Namurian also.'' Indeed, species of *Adiantites* (*A. antiquus* and *A. tenuifolius*) are common in the Pendleian (E1), and rare in the Arnsbergian (E2) of western Europe (Wagner, 1984). We regard the Mauch Chunk sample of *A. antiquus* as one of the rare, younger (Namurian) occurrences of the species, because of its position in a continuous sequence, and its proximity to the base of the type section of the Pottsville Formation, at the Mississippian-Pennsylvanian boundary.

Other indirect evidence for a Namurian (Arnsbergian) age for the top part of the Mauch Chunk Formation comes from the continuous section across the Mississippian-Pennsylvanian boundary along the West Virginia–Virginia border, where the U.S. Geological Survey (USGS) proposed the Pennsylvanian System stratotype (Englund et al., 1979). Here the name Mauch Chunk Group is used (Arkle et al., 1979) for the Late Mississippian sequence encompassing (in ascending order) the Bluefield Formation, Hinton Formation, Princeton Sandstone, and Bluestone Formation. The upper (Bramwell) member of the Bluestone Formation consists of marine beds characterized by conodonts of the *Adetognathus unicornis* Zone (Repetski and Henry, 1983), and brachiopods of the *Orthotetes* n. sp. B–*Diaphragmus* n. sp. A Assemblage Zone (Henry and Gordon, 1992). This fauna permits a correlation with the Imo Formation of northern Arkansas, and the Grove Church Shale at the top of the Chesterian Series in the Illinois basin (Henry and Gordon, 1992). Where the Bramwell Member interfingers with the basal beds of the Pocahontas Formation, both units contain the oldest of Read and Mamay's (1964) Pennsylvanian floral zones (Zone 4, *Neuropteris pocahontas* and *Mariopteris eremopteroides*). Hence, this biostratigraphic relationship parallels that in the Pottsville area, Pennsylvania, and supports a Namurian (Arnsbergian) age for the top part of the Mauch Chunk Formation.

Below Zone 4, the Bluestone Formation also contains a megaflora that Read and Mamay (1964) tentatively assigned to their floral Zone 3 (Late Mississippian zone of *Fryopsis* and *Sphenopteridium*). This megaflora is now referred to a distinct intermediate floral zone (Zone 3A; Pfefferkorn and Gillespie, 1981, 1982) and is separated from floral Zone 3 in the lower Bluefield Formation and upper Greenbrier Limestone (regarded as Mauch Chunk by Read and Mamay, 1964) by an interval of sparse plants encompassing the Princeton Sandstone, Hinton Formation, and most of the Bluefield Formation. As floral Zone 3 broadly correlates with the *Lyginopteris bermudensiformis–Neuropteris antecendens* Zone of Wagner (1984), which ranges from late Visean (III) into the earliest Namurian (E1), it follows that floral Zone 3A is Namurian, and may be as young as Arnsbergian (E2). Although *Adiantites antiquus* has not been recorded from Zone 3A, its presence in the top of the middle member of the Mauch Chunk Formation in the Schuylkill Gap section suggests that it should be correlated with this zone, rather than with Zone 3.

Henry and Gordon (1992) addressed the problem of the relationship of the USGS-recommended Pennsylvanian boundary stratotype at the base of the lower sandstone member of the Pocanhontas Formation, near Bluefield, Virginia–West Virginia (Englund et al., 1985) with the Middle Carboniferous boundary, as accepted by the International Union of Geological Sciences Subcommision on Carboniferous Stratigraphy. This boundary, defined by the first appearance of the conodont *Declinognathodus noduliferus*, is taken for our purposes (Fig. 18) to approximate with the USGS-recommended Pennsylvanian boundary stratoptype.

DiVenere and Opdyke (1991b) suggested a correlation based on magnetic stratigraphy of the Upper Mauch Chunk Formation with the lower part of the Maringouin Formation of New Brunswick. As the boundaries between the Mauch Chunk and Pottsville Formations in Pennsylvania, and the Maringouin and Shepody Formations in New Brunswick are within continuous depositional sequences, our study agrees with the alternative magnetostratigraphic correlation suggested by DiVenere and Opdyke (1991b) in the caption of their Figure 6., i.e., the Upper Mauch Chunk Formation should be correlated with the upper part of the Maringouin Formation (Fig. 18).

ACKNOWLEDGMENTS

We thank Samantha Mackinlay and Xiang Wang for their assistance with section measuring, M.T.D. Wingate, S. Edgecombe, and A. Watson for help with sampling, and K. Huang and R. Enkin for assistance in data analysis. John deLaeter and David Nelson kindly provided access to the Curtin probe. We thank John Stamatakos and Roberto Molino-Garza for excellent reviews that helped to improve this paper. Jon Claoué-Long's contribution is published with the permission of the Executive Director of the Australian Geological Survey Organisation. Geological Survey of Canada publication no. 1997509.

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MANUSCRIPT RECEIVED BY THE SOCIETY DECEMBER 29, 1997
REVISED MANUSCRIPT RECEIVED APRIL 19, 1999
MANUSCRIPT ACCCEPTED OCTOBER 22, 1999

Printed in the USA